

Research Paper

Influence of observed seasonally varying composition on Titan's stratospheric circulation

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ABSTRACT

Titan's atmosphere exhibits variations in composition as it progresses through its seasons. Past observations have shown a substantial enrichment in short-lived molecules in the winter stratosphere above 100 km. Seasonal variations in Titan's stratospheric dynamics also lead to a transient detached haze layer (DHL) above 400 km. The seasonal variations in aerosol opacity and molecular abundance lead to varying radiative heating rates in both the shortwave and longwave spectral regions. In this paper, we report on the effects of a new dataset for aerosol opacity and trace gas abundance derived from Cassini observations on simulations of Titan's stratospheric dynamics with the Titan Atmospheric Model (TAM). We find that including seasonally varying radiative species (SVRS) decreases the autumn and winter polar stratopause temperature by up to 10 K poleward of 60°. We also find a similar increased seasonality in the zonal winds with the early autumn polar jet strengthening by 60 m s⁻¹, while the mid winter jet is unaffected. While including the observationally derived SVRS dataset increases the strength of the seasonal variations of the polar jet, it does not substantially affect the timing of the onset or dissipation of the jet or other seasonal phenomena.

1. Introduction

Titan is the only natural satellite in our Solar System that possesses a substantial, persistent atmosphere, and is the only known body other than Earth to exhibit stable surface liquid. Titan's atmosphere is composed primarily of N₂ (95% in the troposphere, 98% in the stratosphere), with a secondary species of CH₄ (about 4% in the troposphere, 1% in the stratosphere), and trace amounts of H₂, CO₂, and higher-order hydrocarbons and nitriles (i.e., C_xH_y and C_xH_yCN) (Niemann et al., 2010; Hörst, 2017; Mathé et al., 2020). The low surface temperature (about 92 K; Jennings et al., 2019) on Titan enables a methane-based tropospheric hydrologic cycle, evidenced by methane-based lakes, rivers, and rain (Lunine and Lorenz, 2009; Mitchell and Lora, 2016; Hayes, 2016; Hörst, 2017; Hayes et al., 2018). These surface features, however, are shielded from optical observations by Titan's characteristic photochemical haze (also referred to as aerosol) layers.

The Cassini spacecraft orbited Saturn from 2004 through 2017. During its 13.6 year mission, Cassini observed Titan through nearly two full seasons, northern winter through northern spring (southern summer through southern autumn). Over this period, Titan exhibited distinct seasonal transitions in atmospheric composition, temperature, and dynamics (Vinatier et al., 2007, 2015; Teanby et al., 2019; Mathé et al., 2020; Sharkey et al., 2021). Short-lived molecules (such as

HCN and C₃H₄) displayed a substantial enrichment near the winter pole (Teanby et al., 2009; Coustenis et al., 2018; Vinatier et al., 2015; Lombardo et al., 2019; Mathé et al., 2020); in late southern autumn during February 2016, the abundance of HCN at 400 km at 84°S was about 6 × 10⁻⁶, whereas the abundance of HCN at the equator was about 6 × 10⁻⁷ (Mathé et al., 2020). Varying degrees of winter polar enrichment have been observed for virtually all of the trace molecules detected in Titan's stratosphere, including C₂H₆, C₂H₄, C₂H₂, C₃H₄, C₄H₂, and HCN (Mathé et al., 2020). Polar enrichment is likely caused by a meridional overturning circulation (MOC) in the stratosphere that delivers air from the upper stratospheric and mesospheric summer latitudes to the winter polar middle stratosphere (Lebonnois et al., 2001; Crespin et al., 2008; Teanby et al., 2009, 2017).

During most of Titan's year, the stratospheric meridional circulation consists of a pole-to-pole overturning circulation cell that has an upwelling branch above the summer pole and a downwelling branch above the winter pole (Teanby et al., 2012). Near equinox, the upwelling center moves equatorward, passing the equator near equinox, and moving into the opposite hemisphere shortly after. The MOC has been extensively modeled in general circulation models (GCMs) (Hourdin et al., 1995; Newman et al., 2011; Lebonnois et al., 2012; Larson et al., 2015; Lora et al., 2015), and while there are no direct measurements of meridional winds in the stratosphere, the net results

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of the MOC can be inferred from observables, notably the existence of the detached haze layer (DHL) in the summer latitudes (Teauby et al., 2012; Vinatier et al., 2015; West et al., 2018; Mathé et al., 2020; Seignovert et al., 2021), the meridional trends in molecular abundance (Teauby et al., 2012), and the anomalously warm winter stratopause (Achterberg et al., 2008, 2011, 2014). In fact, the warmest part of Titan's stratosphere is the autumn and winter polar stratopause. Even though the winter stratopause is in polar night and is radiatively cooling, adiabatic warming from the descending branch of the MOC is significant enough to cause a net warming (Achterberg et al., 2008, 2011, 2014). Achterberg et al. (2008) estimated an order-of-magnitude downward velocity of 0.5 mm s^{-1} above the northern winter stratopause.

The ascending portions of the MOC are evident in the summer middle stratosphere, a region depleted of aerosol particles. Aerosol production is initiated in the ionosphere (above about 1000 km), where the products of N_2 and CH_4 photodissociation undergo a series of ion reactions to form small spherical particles (Lavvas et al., 2014; Hörst, 2017). Deeper in the atmosphere, close to 600 km, the aerosols grow by the aggregation of several particles, until around 500 km, where surface chemistry may alter the composition and morphology of the particles (Lavvas et al., 2011). Below the 450 km level, the aerosols are increasingly influenced by the atmospheric dynamics, which play a crucial role in the observed distribution of haze particles (Rannou et al., 1997, 2002; Hourdin and Lebonnois, 2004; Crespin et al., 2008; Lebonnois et al., 2012; Larson et al., 2014, 2015). In the summer stratosphere, the ascending motion of the MOC is strong enough to stall the descent of the aerosols, and remove them from the stratosphere, resulting in a region depleted of aerosols and a strengthened DHL above. The depleted zone returns to the normal state by autumn, as the MOC reverses direction (West et al., 2018; Seignovert et al., 2021).

Radiative heating in Titan's atmosphere is controlled by absorption and scattering of solar energy in the visible spectrum by aerosols and molecules (mostly CH_4) that are radiatively active at those wavelengths, and thermal emission and absorption from aerosols, molecules that are radiatively active in the infrared spectrum (hydrocarbons and nitriles), and collision-induced absorption from $\text{N}_2\text{-N}_2$, $\text{N}_2\text{-CH}_4$, $\text{N}_2\text{-H}_2$, and $\text{CH}_4\text{-CH}_4$ pairs (Tomasko et al., 2008; Bézard et al., 2018). In most of the stratosphere, the radiative heating is comprised of the absorption and emission from haze and molecules, with collision-induced absorption only playing a significant role in the troposphere and lower 50 km of the stratosphere. The seasonal evolution of both molecular abundance and aerosol opacity should therefore play a dominant role in Titan's stratospheric radiative balance.

1.1. Previous simulations of Titan's stratosphere

The first fully three-dimensional model to accurately recreate the general characteristics of Titan's atmosphere was that of Hourdin et al. (1995). The model assumed a constant and meridionally uniform distribution of trace gases and aerosols, with only the solar insolation varying with season. While the general structure and seasonality of the atmosphere were simulated well, the magnitudes of the meridional temperature gradients and zonal wind were significantly weaker than those observed by the Voyager 1 and Cassini spacecraft (Flasar et al., 1981, 2005; Flasar and Achterberg, 2009; Achterberg et al., 2008), with a simulated jet strength of 120 m s^{-1} at the northern winter solstice. Unfortunately the vertical extent of the jet was unable to be resolved as the model top was too low at 8 Pa. In the three-dimensional TitanWRF model (Newman et al., 2011), which used a higher model top at 0.1 Pa (420 km) the simulated stratospheric winds reached a maximum speed near 180 m s^{-1} in mid winter, though the stratopause temperatures often exceeded the 180 K observed by the Composite Infrared Spectrometer (CIRS) on Cassini (Achterberg et al., 2008, 2011). While the model exhibited accurate seasonality, it still assumed a constant and uniform distribution of trace gases and aerosols.

To address the affects of varying aerosol opacity on Titan's atmosphere, the three dimensional Institut Pierre Simon Laplace (IPSL) GCM (Lebonnois et al., 2012) included a haze microphysical model that was coupled to the model dynamics. Their simulations successfully produced a detached haze layer, though it was about 150 km lower than observed by Cassini. Additionally, a trait unique to this model was its ability to create a local zonal wind minimum at a similar altitude as the dramatic minimum observed by Huygens in the lower stratosphere (Bird et al., 2005). The authors attributed this minimum to the transitional nature of the level between the sub-annual radiative relaxation time of lower pressures, and the longer relaxation time of higher pressures. While this model captured the magnitude of zonal wind throughout most of the stratosphere well, it lacked realism in the upper stratosphere, and the simulations did not exhibit a distinct stratopause, with temperatures of nearly 240 K near 1 Pa compared to the observed 150 K. An additional three-dimensional GCM with coupled haze microphysics was published in Larson et al. (2015), building on the model of Friedson et al. (2009), which, while successfully simulating Titan's thermal structure through 1 Pa, was unable to generate Titan's strong prograde polar jet. The inability of the Friedson et al. (2009) model to produce strong zonal winds prompted Larson et al. (2014) to include an unphysical forcing of the equatorial zonal winds. With this forcing, the model adequately simulated the evolution of Titan's detached haze layer, though at about 100 km lower than observed by Cassini (West et al., 2011). The authors also found very little meridional variation in haze opacity below the 100 Pa level (with the exception of a slight decrease near 10^4 Pa at latitudes equatorward of 50°).

The effects of seasonal variations in atmospheric chemistry were also studied by several two-dimensional models. Rannou et al. (2004) (based on Hourdin et al. (1995)) included a haze production scheme that generated and transported haze based on microphysics and dynamics. Inclusion of their haze microphysics strengthened the simulated jet by nearly 40 m s^{-1} , with their equinox and solstice polar jets exceeding 200 m s^{-1} . Building on this, Hourdin and Lebonnois (2004) showed that coupling molecular production rates from Lebonnois et al. (2001) had no significant effects on the stratospheric zonal wind. The effects of tracer advection by the global circulation were addressed in an additional update in Crespin et al. (2008). The model included a haze microphysical parameterization that consistently generated aerosol particles at 400 km. Production and depletion of trace molecules were parameterized in the stratosphere, with fluxes of species from above the model top varying with latitude and season. Even though this model inherently included uncertainties due to its two-dimensional nature, it reproduced the adiabatic warming of the polar stratopause, the magnitude of zonal winds, and polar enrichment of trace molecules. However, as with the simulated temperatures from many other Titan GCMs, the northern mid winter stratopause was significantly warmer (240 K) than observed by Cassini (200 K).

The Titan Atmospheric Model (TAM; Lora et al., 2015) is a three-dimensional GCM that has previously been used to study the middle and lower atmosphere of Titan, and in particular the methane hydrologic cycle (Lora et al., 2015; Lora and Ádámkovics, 2017; Faulk et al., 2020). The model configuration presented in Lora et al. (2015) extended from the surface to 4 Pa (about 420 km). The simulations recreated Titan's lower atmosphere in agreement with HASI observations; however, compared to CIRS measurements, the stratosphere tended to be too warm by about 10 K, and the stratopause was too high by 50 km. The zonal winds reached a maximum of about 150 m s^{-1} in mid autumn, lower than the 200 m s^{-1} inferred from meridional temperature gradients measured by CIRS (Achterberg et al., 2008). As with the other three-dimensional models, previous versions of TAM did not include seasonally varying radiative species.

Building on the above progress, in this paper we report on a new data product developed from Cassini observations and ensuing TAM simulations that take into account seasonal variations of radiatively

active species (hereafter, “seasonally varying radiatively active species” will be abbreviated SVRS). Using this data product, we are able to accurately simulate the effects of Titan’s seasonally varying atmospheric composition on the general circulation of Titan’s atmosphere. Previous studies have hypothesized on the effects of Titan’s strong compositional seasonality, especially near the poles (Teanby et al., 2008, 2017), but this has not been explicitly simulated previously using a three-dimensional GCM. The first component of our SVRS data product includes information on the abundance of C_2H_6 , C_2H_4 , C_2H_2 , C_3H_4 , C_4H_2 , and HCN derived from meridional and vertical measurements from CIRS (Teanby et al., 2019; Mathé et al., 2020). The second includes meridional and vertical information on cumulative aerosol opacity from the surface through 500 km, derived from observations from the Cassini Image Science Subsystem (ISS) Narrow Angle Camera (NAC) in the ultraviolet region above 400 km (Seignovert et al., 2021), and the Huygens Descent Imager/Spectral Radiometer (DISR) below 200 km (Doose et al., 2016).

2. Model

TAM is based on the Geophysical Fluid Dynamics Laboratory’s (GFDL) Flexible Modeling System (FMS) spectral dynamical core (Gordon and Stern, 1982). The primitive equations are solved horizontally with prognostic variables represented as spherical harmonics and vertically through a finite difference method. The vertical coordinate is a hybrid sigma coordinate, which transitions gradually to be identical to pressure above around 100 hPa. A physics package is coupled to the dynamical core to include effects such as diabatic heating from radiation, moist convection, and surface interactions. A thorough description of each physics parameterization is given in Lora et al. (2015), and several other updates are discussed elsewhere (Lora and Ádámkóvics, 2017; Faulk et al., 2020; Battalio et al., 2022); here we discuss our recent improvements to the model stratosphere. A note of clarification regarding the language of this paper: we use “below” to refer to lower in altitude and “above” to mean higher in altitude, unless otherwise noted. For example, “below 0.6 Pa” indicates the area below the altitude corresponding to 0.6 Pa.

2.1. Radiative transfer

Our updated version of TAM includes a largely rewritten and completely generalized radiative transfer scheme that may be easily modified to vary the optical constants of radiatively active components in the model. Radiative transfer calculations are separated into solar (2000–40,000 cm^{-1} , $<5 \mu m$) and thermal-infrared (1–2000 cm^{-1} , $>5 \mu m$) spectral regimes. Fluxes in each regime are calculated making use of a nongray, multiple scattering, plane parallel two stream approximation from extinction optical depths, single scattering albedos, and asymmetry parameters, as in Lora et al. (2015). Insolation is varied according to seasonal and diurnal cycles, and therefore includes polar night at appropriate latitudes.

Opacity from molecular absorption is calculated using correlated- k coefficients (Lacis and Oinas, 1991) determined from spectral line data compiled in the HITRAN database (Gordon et al., 2017). Reference tables for the opacity of each molecule are calculated before runtime across a range of temperatures and pressures. The correlated- k coefficient tables are then read by the model and opacities for the varying atmospheric compositions are calculated at runtime. Absorbers include CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , C_3H_4 , and HCN. The abundance of each absorber can be prescribed using either a static profile across all latitudes and times—as was used in previous TAM versions—or (with the exception of CH_4) a “climatological” profile which varies in latitude, altitude, and solar longitude (season), as derived from Cassini observations in the SVRS dataset.

Opacity from collision-induced absorption (CIA) is calculated for N_2 - N_2 , N_2 - CH_4 , N_2 - H_2 , and CH_4 - CH_4 pairs from constants included in

the HITRAN database (Gordon et al., 2017). CIA is only calculated for the infrared band, not the visible band, since effects from this process are negligible at wavenumbers greater than about 600 cm^{-1} .

Solar wavelength optical properties (opacity, single scattering albedo, and asymmetry parameter) from Titan’s atmospheric aerosol are derived from the Doose et al. (2016) model, which was based on observations from the Huygens probe. Optical depths at wavelengths shorter than 0.5 μm (20000 cm^{-1}) are extrapolated with a power law fit, assuming that the haze becomes more opaque with decreasing wavelength. Values for the single scattering albedo and asymmetry parameter are extrapolated at a constant value beyond the bounds of the DISR data. While the extrapolation to higher wavenumbers is an unavoidable model assumption, most of the solar energy deposited on Titan is contained within the DISR band. In the thermal infrared, the aerosol is assumed to be perfectly absorbing, thus both the single scattering albedo and asymmetry parameter are identically zero for all wavelengths. Opacity in this region is calculated from volume extinction coefficients from Cassini CIRS observations from 20–1500 cm^{-1} (Vinatier et al., 2010; Anderson and Samuelson, 2011). Between 1500 cm^{-1} (the upper limit for CIRS observations) and 2000 cm^{-1} (the edge of the solar wavelength regime), opacity is interpolated using a power law fit. The model should not be sensitive to the interpolation, because at these wavelengths very little energy is transmitted. The extinction coefficients are assumed to be vertically constant between 0 km and 80 km, and decrease with a scale height of 65 km above 80 km.

In addition to these static aerosol profiles, we also generated a dataset of seasonal variations of haze opacity (included in the SVRS dataset), which combines the Doose et al. (2016) model from Huygens DISR measurements and seasonally varying opacity in the detached haze layer above 350 km from Seignovert et al. (2021) (described in the following subsection). Combined with the seasonal variations of molecules, the varying opacity dataset enables our version of TAM to capture the seasonal variations in radiative forcing in Titan’s stratosphere.

2.2. Improvements to upper level stability

Previous versions of TAM were optimized for simulating Titan’s lower atmosphere, with the stratosphere either not fully resolved or simulated at a low vertical resolution. As our aim here is to more robustly study the behavior of Titan’s stratosphere, we necessarily have to utilize a higher vertical resolution and higher model top. Unfortunately, the stability of a GCM tends to degrade as both the model top is raised and vertical resolution is increased. The former is caused by the decrease in pressure as altitude is increased, which leads to vertically propagating waves growing unrealistically large near the model top. The latter is limited by the model time step, with shorter time steps (and longer run times) necessary to accurately simulate the higher resolution atmosphere. To improve stability at the model top we apply several wind and temperature dampings. Similar to previous versions of TAM, we apply a Laplacian (∇^2) sponge to the zonal mean and eddy components of the zonal and meridional winds. In this updated model, we also include an energy-conserving Rayleigh damping of the eddy component of the zonal and meridional winds with a strength (inverse timescale) that varies vertically with a maximum at the model top and disappearing by 0.5 Pa. We also adapt the Kelvin–Helmholtz (K-H) wave parameterization from the FV3 dynamical core (Harris et al., 2021), which identifies grid points where the local Richardson number between two pressure levels, $Ri = \frac{g}{\theta} \frac{d\theta}{dz} / (\frac{du}{dz})^2$, with g the gravitational acceleration, θ the potential temperature, and u the total horizontal wind, is less than a threshold value, Ri_{thresh} . Where $Ri < Ri_{thresh}$, the zonal and meridional winds as well as dry energy are mixed between layers with an efficiency equal to $1 - \frac{Ri}{Ri_{thresh}}$. Therefore, more unstable regions are mixed more thoroughly. Ri_{thresh} is typically taken to be 0.25, but we increase this value at the upper-most levels to further improve the stability of the model.

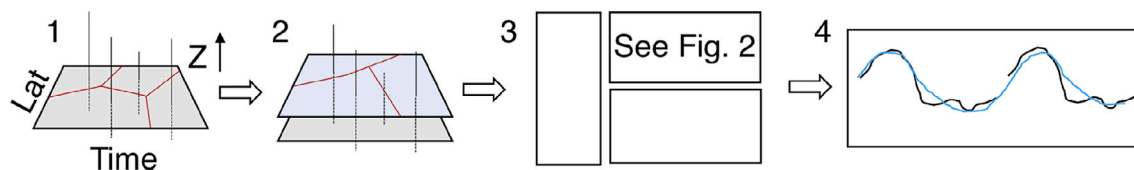


Fig. 1. A schematic diagram of the development of the SVRS dataset from raw data to the final product. The original data were regridded onto a uniform latitude, solar longitude, and pressure grid (1). A Voronoi tessellation was then performed on the next pressure level, and each region scaled to match the measured profile at the corresponding latitude–time coordinate (2). Each meridional, temporal, and vertical transect was then smoothed (3; see Fig. 2); then, each time series was shifted by 180° , shifted to the opposite latitude, and scaled to match the end points of the original time series (4). Lastly, the trace species were condensed out of the atmosphere and a constant profile set below (not shown).

2.3. Model configuration

TAM was run with the updated radiative transfer scheme described in Section 2. The simulation included 50 atmospheric layers, extending from the surface to 16 scale heights (about 10^{-2} Pa). A diffusive sponge was applied to the top layer of the model to minimize wave reflections and increase numerical stability. The surface was initialized with the wetlands scenario used in previous versions of TAM (Lora and Ádámkovics, 2017) with plentiful surface liquid methane at latitudes poleward of 60° in both hemispheres. While this is less realistic than more recent previous versions of TAM (Faulk et al., 2020; Battalio et al., 2022; Lora et al., 2022), the wetlands scenario captures the essential aspects of the tropospheric methane cycle and is substantially faster to run.

To save computation time during the spin-up phase, the initial rest-state atmosphere was nudged towards a previously spun-up simulation of TAM, that was interpolated to a higher vertical resolution. When the atmosphere was sufficiently spun-up, the nudging was disabled and the model was allowed to integrate for an additional few Titan years to allow any instabilities caused by the nudging to dissipate. Using this spun-up state as a starting point, the model was run for 10 Titan years using static molecular and aerosol profiles representative of the average of the SVRS dataset. From the same starting point, TAM was run also for 10 years including SVRS. The two model runs—the SVRS and static runs—are compared throughout this paper. Unless otherwise stated, all figures use data from the SVRS simulations. To study the effects of SVRS on the simulated atmosphere without complications from interannual variability, each of the 10-year simulations was averaged to yield a single statistically representative year of data.

2.4. Seasonal variations of molecules

To generate a product of seasonally varying trace gases on a uniform grid that includes vertical, meridional, and temporal structure, we combined measurements at the 100 Pa level from 2004–2017 (Teanby et al., 2019) with vertical profiles from limb measurements observed at discrete latitudes at non-uniform times (Mathé et al., 2020). We used a grid with a resolution of 99 vertical levels, equally spaced in log-pressure from the surface (1.465×10^5 Pa) to 10^{-3} Pa, 1° in latitude, and 1° L_s in time (L_s is the solar longitude, a measure of Titan’s season, with northern spring equinox, summer solstice, autumn equinox, and winter solstice occurring at $L_s = 0^\circ, 90^\circ, 180^\circ$, and 270° respectively).

Meridional variations of trace molecules are based on the time-latitude maps at the 100 Pa level (Teanby et al., 2019) that were determined through modeling CIRS nadir observations across Titan’s disk. While the 100 Pa maps include nearly complete coverage between 2004 and 2017, there are a few periods during which CIRS was unable to perform the necessary nadir observations, resulting in gaps in the data coverage. We interpolated the published maps to fill gaps in data, and extrapolated outward to fill the edges of the map. A complete overview of the vertical observations of trace molecules in Titan’s atmosphere was published in Mathé et al. (2020). Measurements were made by modeling limb spectra of Titan between 100 km and 500 km. We interpolated all profiles onto a common pressure grid.

After the initial regridding, we vertically extrapolated 100 Pa-level data first by duplicating the 100 Pa map at a lower pressure level (upward in altitude). Then we used a non-overlapping Voronoi mesh to determine the time–latitude map regions corresponding to the nearest available limb measurement (red lines, Fig. 1 panel 1). We calculated the Voronoi mesh separately at each pressure level to ensure that vertical profiles with differing pressure extents were included only for pressure levels where they include data. Therefore, at the highest and lowest pressures, there were only a few very large Voronoi regions, corresponding to only the profiles with highest or lowest extents. In each region, we applied a multiplicative scaling factor to match the abundance in the map at the location of the profile to the measured profile abundance. We applied the scaling factor to the entire Voronoi region to retain the sub-region scale meridional and temporal variation observed in the 100 Pa map. We repeated this process for each pressure level, with the time–latitude map being duplicated from the previously calculated pressure level (Fig. 1 panel 2). This extended upwards to 1 Pa, and downwards to the surface. If no vertical profiles existed at a given pressure level (usually below 10^3 Pa and above 10^{-2} Pa), the previous level was duplicated with no scaling. The final product of this stage was a completely populated three-dimensional cube in latitude/time/pressure that represented nearly half a Titan year. We then applied a Gaussian filter to each vertical profile, with a half-width at half-maximum (σ) of approximately 1 scale height, and truncated at 2σ , a Gaussian filter to each meridional transect with $\sigma = 7^\circ$ and truncated at 2σ , and a Gaussian filter to each time series with $\sigma = 1$ Earth year and truncated at 2σ , such that information from 4 Earth years (about 54% of a Titan season) was included at each point. We chose these σ values as they represent a balance between sufficiently smoothing the data and retaining information on the climatological (i.e., average large scale) seasonal variations in Titan’s atmosphere (smoothing schematically shown in Fig. 1 panel 3, and shown for HCN in Fig. 2).

Titan’s atmosphere is expected to display some degree of seasonal symmetry. We would expect Titan’s mid latitude winter to behave similarly whether it is in the northern hemisphere or the southern hemisphere. This has been observed to some degree, as trace species are known to be enriched in high-latitude winter due to the inferred meridional overturning circulation (Coustenis et al., 2018; Teanby et al., 2019; Mathé et al., 2020). While this approximation has its flaws (Titan has a non-zero effective eccentricity around the Sun), it is a decent first order estimate given the limitations of the data we have. To extrapolate the <0.5 Titan year observations to an entire Titan year, we translated information from one hemisphere to the other: to complete the dataset for the half of the Titan-year when there are no CIRS observations, we copied individual time series to the opposite latitude and shifted them by 180° . This effectively duplicated the seasonal trend at each latitude to the trend during the same season in the corresponding latitude of the opposite hemisphere. We interpolated linearly over the gap between the Cassini-era and non-Cassini-era time series, and smoothed the abrupt transition with a moving average (schematically shown in Fig. 1 panel 4).

The final step in producing the SVRS dataset was to account for saturation of the molecules in the lower atmosphere. We calculated saturation curves for each molecule with Antoine coefficients available

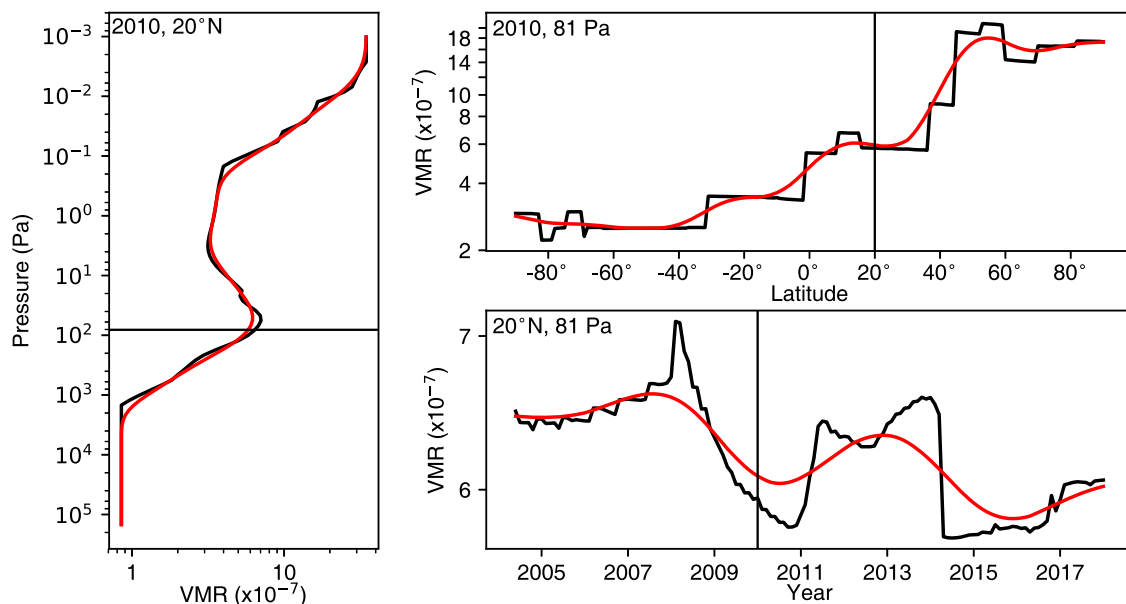


Fig. 2. Comparison between the initial HCN abundance trends from the Voronoi tessellation (black) and HCN abundance trends after smoothing is applied (red). Horizontal and vertical black lines indicate the pressure, latitude, and/or year corresponding to the other two panels. The vertical smoothing is applied first, followed by meridional smoothing, and, lastly, temporal smoothing. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

from the National Institute of Standards and Technology (NIST) webbook and temperature informed by [Teanby et al. \(2019\)](#). At altitudes below the condensation level (typically in the lower stratosphere) we assumed a constant vertical profile.

2.5. Seasonal variations of aerosol opacity

Due to the nature of the DHL opacity observations presented in [Seignovert et al. \(2021\)](#), measurements from individual flybys of Titan are available as 2-dimensional meridional-vertical transects. As each observation had a unique viewing geometry, the meridional extents of the observations are not uniform. Additionally, the vertical extent of the data are non-uniform, with some observations becoming unreliable below about 400 km. The observations were also not performed at a constant cadence, with some periods of time including several observations for a given latitude and other periods containing none. Similar to the method used for assimilating the molecular compositions, a robust interpolation method was needed to generate a uniform dataset of seasonally varying aerosol opacity.

We initially interpolated data from [Seignovert et al. \(2021\)](#) onto a uniform altitude grid and smoothed them vertically with a Gaussian filter with $\sigma=3$ km. Stepping through time then through altitude, we began by extrapolating the meridional transect of each observation. If the latitude bounds of the observations were within 30° of either pole, we extrapolated the meridional transects at a constant value; otherwise, we left the polar data blank, and we extrapolated data at the latitude bounds at a constant value for 10° . Then, we applied a Gaussian smoothing to the meridional transect with $\sigma=5^\circ$. Once the meridional transect smoothing was completed, we followed a similar procedure and interpolated each across a single latitude and altitude to produce a continuous time-series. We then smoothed each meridional transect again with $\sigma=10^\circ$, and smoothed each time series with $\sigma=2$ L_y . Finally, we smoothed each vertical profile with $\sigma=3$ km. We determined the σ values used to smooth the data to be the smallest values that sufficiently removed small-scale variations in the observations, as well as discontinuities due to the interpolation, while also retaining large-scale characteristics. At this point, we sought to convert the DHL opacity from altitude-space to pressure-space, to be more easily assimilated by the GCM. Using CIRS-derived temperatures from [Teanby et al. \(2019\)](#),

we converted the altitude grid used in the existing array to pressure, then interpolated each profile to a uniform pressure grid.

With the DHL interpolation complete, we sought to incorporate haze characteristics in the lower atmosphere as calculated in [Doose et al. \(2016\)](#). As shown in [Larson et al. \(2015\)](#), aerosol opacity is not expected to vary significantly in latitude or time below the 100 Pa level, though modest variations on short timescales have been observed ([Nichols-Fleming et al., 2021](#)). Therefore, we set the opacity below this level to be equal to the vertical profile described in [Doose et al. \(2016\)](#). To incorporate the DHL measurements, we first removed unrealistic opacity values from the initial DHL interpolation described above, including: a) τ below 0.6 Pa, where the opacity measurements tended to be less reliable ([Seignovert et al., 2021](#)), b) where the interpolated τ exceeded unity, and c) where τ exceeded the opacity calculated at the top of the [Doose et al. \(2016\)](#) model to ensure a decreasing opacity with altitude. We then interpolated over the region below 0.6 Pa and above 100 Pa (between the DHL and static profiles from [Doose et al. \(2016\)](#)) to include the seasonal effects of the upper altitudes, while being constrained by the static nature of the lower altitudes. Finally, we extracted information on the opacity spectrum of the [Doose et al. \(2016\)](#) model at each pressure level, and normalized this spectrum relative to the opacity at 22586 cm^{-1} (the wavenumber at which the DHL observations were performed). We then used the normalized [Doose et al. \(2016\)](#) spectrum as a multiplier to determine the aerosol opacity at all wavelengths in the model for all latitudes and times.

3. Results

3.1. Results from the SVRS dataset

A comparison between eight observed HCN profiles from [Mathé et al. \(2020\)](#) and contemporaneous profiles from the SVRS dataset is shown in [Fig. 3](#). The Voronoi tessellation at 10 Pa is shown in the left column. Each region corresponds to a single profile, indicated by a black dot. The eight “test” profiles shown to the right are indicated by red circles on the tessellation. Unsurprisingly, our SVRS dataset captures the variability in both space and time measured across different profiles well (since the SVRS dataset is driven completely by the observations). Some profiles (2016.2 at 84°S and 2012 at 79°N) show

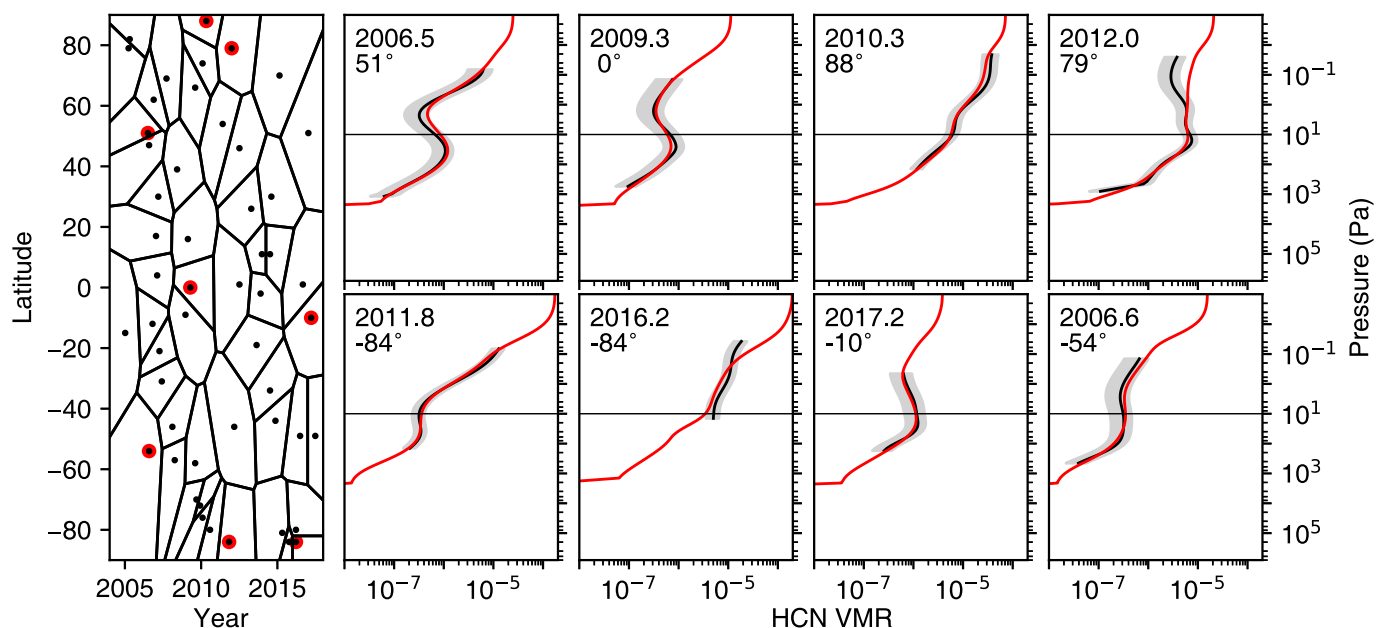


Fig. 3. (Left) Voronoi tessellation of the 10 Pa level (black lines) and locations of Mathé et al. (2020) HCN observations (black dots). (Right) Comparisons between observed HCN profiles (black profiles, indicated by red dots on the tessellation) and contemporaneous SVRS profiles (red). Uncertainties from observations are shaded gray. The pressure of the Voronoi tessellation is indicated by a solid black line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

deviation from the shape of the observed profiles. This is caused by nearby observations with different measured abundances, which lead to deviations between SVRS and the observations due to the smoothing process. The smoothing of small scale variations, while producing imperfect matches to the observed profiles, allows the SVRS dataset to capture large scale behavior while ignoring small scale noise. This allows for smooth forcing of the GCM.

The abundance cross-section for HCN at each solstice and equinox, as included in SVRS, is shown in Fig. 4. Clearly visible is the progression of polar enrichment over the course of a year. In the high-latitude summer hemisphere ($L_s=90^\circ$) there is little-to-no meridional gradient in HCN abundance. A uniformly higher abundance above 400 km at all latitudes begins to develop in the autumn hemisphere ($L_s=180^\circ$). By winter ($L_s=270^\circ$), there is a full polar enrichment, with a steep meridional gradient in abundance at all altitudes above 100 km, as well as a higher abundance at higher altitudes. Finally, in the spring hemisphere ($L_s=0^\circ$) the relaxation of the meridional gradient back to the summer state begins.

3.2. GCM results

3.2.1. Atmospheric temperatures

Fig. 5 shows the simulated zonal mean temperature in northern mid winter ($L_s=300^\circ$), early spring ($L_s=10^\circ$), late spring ($L_s=60^\circ$), and summer solstice ($L_s=90^\circ$). At $L_s=300^\circ$, the temperature of the winter stratopause exceeds 195 K and is located near 10 Pa, which is both warmer and higher than the stratopause at all other latitudes (175K–185K at 50 Pa). Below the winter stratopause, the stratosphere is up to 20 K cooler than at the equatorial and summer latitudes. At 1000 Pa, for example, the winter polar temperature is about 140 K compared to the equatorial 155 K. In early northern spring ($L_s=10^\circ$) the stratosphere undergoes several modifications. In the northern hemisphere, the stratopause cools to 185 K while the stratospheric temperature remains the same. In the southern hemisphere, the stratopause pressure decreases to near 3 Pa, while the vertical temperature gradient steepens to reflect the structure at the opposite pole. By mid southern autumn ($L_s=60^\circ$), the south polar stratopause descends slightly to the 10 Pa level, and increases in temperature to above 190 K. This is a similar

pressure and temperature as the mid winter stratopause of the opposite hemisphere (as in the top panel). The cooling of the mid and lower stratosphere also leads to temperatures approximately 20–40 K lower than at other latitudes. This trend continues through the solstice ($L_s=90^\circ$). A comparison to the temperature calculated in Achterberg et al. (2011) for the northern hemisphere at $L_s=4^\circ$ is shown in dashed lines in the second panel. Our temperatures are generally higher throughout the stratosphere, though the extent of the cooling in the high latitudes, as estimated by the slope of the isotherms, is generally consistent. The vertical extent of the stratopause is also comparable to the observations.

Simulated temperatures are compared to temperature profiles from radio occultations (Schinder et al., 2012) in Fig. 6. During mid summer (top row), the simulations are in good agreement with the observations. The main differences are a slightly warm (about 10 K) middle to upper stratosphere, and a descended stratopause (the modeled stratopause is at 30 Pa, the observed stratopause is above 10 Pa). More interesting behavior occurs during autumn and winter, during which time a strong thermal inversion centered at 1000 Pa at latitudes poleward of 74°N was observed. The inversion was not present in latitudes equatorward of 52°N , though the observed winter stratosphere was still substantially colder than at other latitudes. At times corresponding to the radio occultations (late winter), the simulated temperature does not exhibit a significantly cooler stratosphere; however, as shown in Fig. 5, the simulated temperatures are substantially colder in the polar stratosphere beginning in early autumn. The most intense cooling dissipates by mid winter, as shown in Fig. 7. The modeled cooling of the polar stratosphere precedes observations by about half a season, as was the case in Lora et al. (2015).

The top panel of Fig. 7 shows the temperature averaged between 85°S and 60°S at 50 Pa and 3 Pa. The temperature evolution in the mid stratosphere (50 Pa) follows a distinct evolution: (a) a period of weak cooling from mid spring to just before the autumn equinox; (b) a period of rapid cooling beginning immediately before autumn equinox, and extending to about one-third of the way through autumn, where the temperature is lowest; (c) a period of warming from early-autumn through early winter, with the rate of warming decreasing as the temperature rises; (d) a secondary, very weak cooling from mid to late winter; and (e) rapid warming from just before the spring

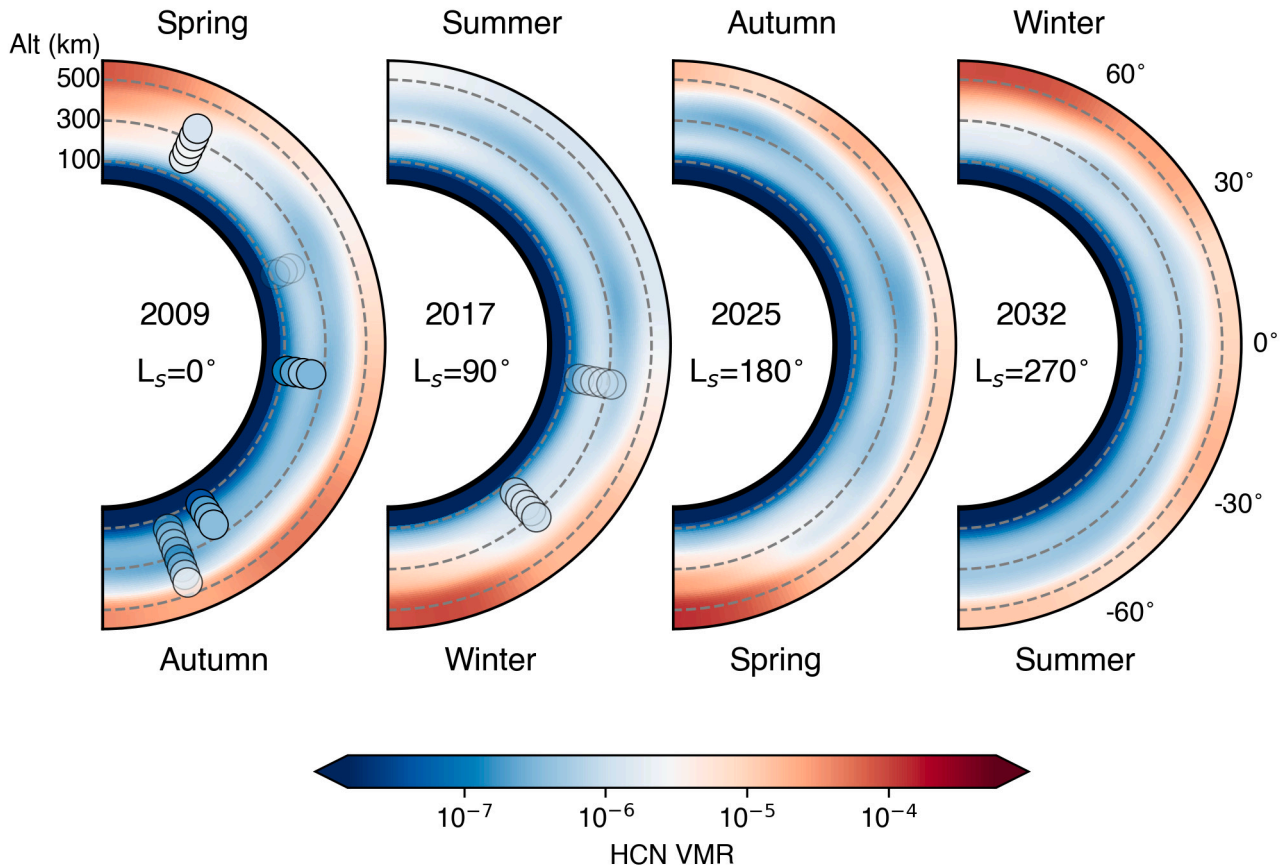


Fig. 4. Seasonal abundance of HCN at four solar longitudes from our SVRS dataset. CIRS measurements from Mathé et al. (2020) are indicated by shaded circles, following the same color scheme. Higher transparency circles represent observations further from the indicated L_s . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

equinox to the temperature maximum described in (a). When SVRS is included, the summer stratospheric temperatures are at most 5 K warmer while the winter temperatures are 5 K cooler. The timing of the maxima and minima are unaffected by SVRS. At higher levels (3 Pa), the timing of temperature changes coincides with lower levels, but the change of temperatures are in opposite directions. During the mid stratosphere cooling from mid spring to late summer, the upper stratospheric temperature only varies by up to 5 K. Inclusion of SVRS plays a diminished role in the global stratospheric temperatures at lower pressures compared to the mid stratosphere; likewise, the timing of temperature changes in the upper stratosphere is largely unaffected. The exception to this is from late summer through early autumn, when the stratopause is the warmest. At this time, the stratopause is 5–8 K warmer when SVRS is included.

Fig. 8 shows the evolution of temperatures across the simulated average Titan year at pressure levels corresponding to the lower mesosphere (1 Pa), the stratopause (11 Pa), mid stratosphere (67 Pa), and lower stratosphere (322 Pa; contours) and the effects of including SVRS (shading). As is evident in Fig. 5, the mid stratosphere is coldest poleward of 60° in mid autumn. The northern high latitude temperature then increases by the beginning of winter, before it returns to a state similar to other latitudes by the following spring. Similarly, the stratopause reaches a thermal maximum in mid autumn, which persists through the beginning of the following spring. Inclusion of SVRS most strongly affects the polar regions during mid autumn, though additional cooling is evident in early spring (for example, northward of 50°N, $L_s=0^\circ$ –60° at 11 Pa). At the highest levels (1 Pa), polar heating begins

about 10° L_s before equinox, and persists for almost 210° L_s , over half a year. The warming is diminished in the first half of winter, then strengthened around spring equinox before cooling back to low-latitude temperatures by mid-spring.

Deeper in the atmosphere, at 11 Pa, warming is marginal until late autumn, at which point the stratopause descends and the 11 Pa level warms to its highest temperature. This persists at over 200 K throughout winter, and gradually returns to the low-latitude temperatures by the middle of spring. At 67 Pa, the mid stratosphere, the polar latitudes are cooler than at lower latitudes, in contrast to the warmer temperatures experienced at higher altitudes. The cooling does not begin until very close to equinox (later than the high altitude warming), and only persists until winter solstice. At solstice, there is a weak meridional temperature gradient across all latitudes, which strengthens by spring equinox. At this point, the temperature gradually warms to summer temperatures. The effects from SVRS are strongest at higher levels, with the autumn stratopause cooled by up to 10 K, and the summer stratopause warmed by up to 5 K. In the mid stratosphere, the effects of SVRS are strongest in the southern hemisphere, with summer (winter) polar temperatures warmed (cooled) by up to 5 K.

At $L_s=303^\circ$, Achterberg et al. (2011) (Fig. 3 panel a of their paper) measured the winter stratopause north of 50°N at 210 K, centered between 1–2 Pa. At lower latitude, the stratopause was deeper, closer to 10 Pa, and cooler, near 185 K. By equinox, their measured stratopause remained near 1 Pa north of 60°N, but descended to 20 Pa south of that latitude. Additionally, the polar stratopause temperature decreased to 180 K to match with the lower latitude stratopause temperature.

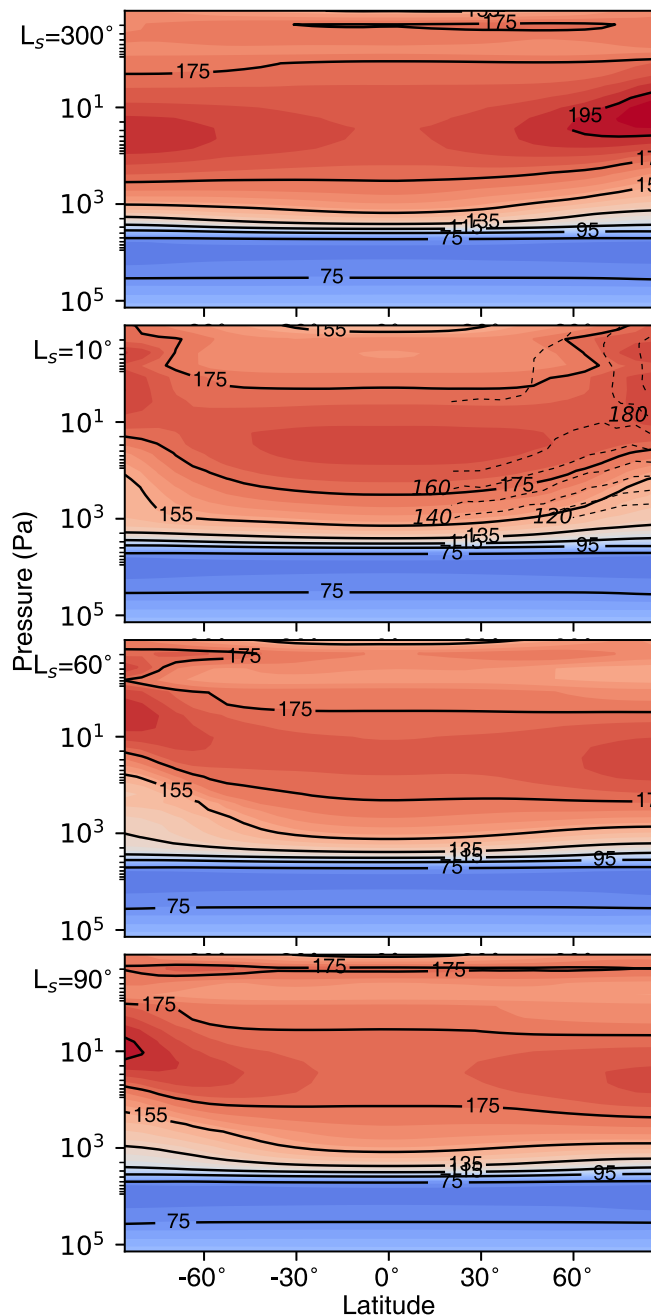


Fig. 5. Simulated zonal mean temperature during northern (southern) mid winter (summer), early spring (autumn), late spring (autumn), and summer (winter) solstice. The dashed lines with italic labels in the second panel indicate the temperatures retrieved in Achterberg et al. (2008) for the T63 flyby near $L_s=4^\circ$.

Our simulated high latitude early winter stratopause is too cool by about 10 K but our mid latitude stratopause is consistent with the observed temperatures. The pressure of our simulated high latitude winter stratopause is also too high (10 Pa). The equinox stratopause we simulate is similar to that observed in Achterberg et al. (2011), with both the decreasing altitude of the stratopause towards the equator and the stratopause temperature consistent with observations, although our high latitude stratopause remains slightly too deep.

3.2.2. Zonal winds

Simulated stratospheric (above 10^4 Pa or about 50 km) zonal-mean zonal wind in southern mid summer, early autumn, late autumn, and

at the winter solstice (roughly the same time-coverage of the Cassini mission) is shown in Fig. 9. (Seasonal variations in the tropospheric zonal winds are orders of magnitude weaker, and are not shown in this figure.) During southern mid summer ($L_s=300^\circ$), the simulated zonal wind is characterized by strong westerlies at all latitudes, with the strongest winds located in the northern (winter) hemisphere. Between 5°S and 60°N , the zonal wind is uniform across a single pressure level, with a maximum wind speed of 130 m s^{-1} at 100 Pa. In the southern (summer) hemisphere, the wind speed gradually decays from the equator towards the pole, with the wind achieving a maximum speed closer to 1000 Pa. The evolution of zonal wind averaged between 85°S and 60°S for the stratopause and middle stratosphere are shown in the bottom panel of Fig. 7. Similar to the evolution of temperature in the same region, the zonal winds show distinct phases of (a) relative constant velocity in summer, (b) rapid increase in early autumn, (c) gradual decrease from late autumn through early winter, and (d) moderate strengthening through early spring.

Ground-based measurements of globally averaged stratospheric winds from this season (2001–2005), summarized in Kostiuk et al. (2006), indicated a maximum zonal wind of $120 \pm 50\text{ m s}^{-1}$ centered at 100 Pa, decreasing with height to 80 m s^{-1} by 1 Pa, and decreasing with pressure to 80 m s^{-1} by 1000 Pa. A direct comparison of our simulated winds to these ground-based observations is hindered by the observations' disk-averaged nature, meaning they are not truly global as they encompass the latitudes facing Earth at the time of observation (the southern summer hemisphere, in this case). Comparing only our low-latitudes and summer hemisphere to the observations, the simulated winds are of a lower magnitude (as with comparisons of our winds to other observations), and the location of the maximum wind is slightly deeper (again consistent with comparisons to other observations) — though the match is better if only the equator is compared. Our simulations and the ground-based observations are in agreement with the general shape of the wind field, however, as both datasets show a rapid decrease in zonal wind above the 100 Pa level. Our simulations also show a secondary jet centered at the equator above 1 Pa with a maximum speed of 60 m s^{-1} .

In early southern autumn ($L_s=10^\circ$), wind speed above 100 Pa drastically increases across all latitudes, while winds below 100 Pa remain largely unchanged. A strong polar jet forms between 30°N and 60°N in the spring hemisphere between 1 Pa and 100 Pa. At these same levels, the zonal wind in the southern (autumn) hemisphere also rapidly increases to a maximum of 120 m s^{-1} , though no well-defined jet is apparent. By $L_s=14^\circ$ (corresponding to early 2011 Cassini observations), the zonal wind field is hemispherically symmetric, with the strengthening south polar jet roughly equal in strength and location to the decaying north polar jet. There is also significantly enhanced zonal flow within 30° of the equator above 1 Pa. This symmetry does not last long, as by $L_s=22^\circ$ (not shown), the north polar jet dissipates. Nearly one half-season later, at $L_s=46^\circ$, the northern hemisphere zonal winds continue to weaken, and the well-defined jet in the southern hemisphere descends and strengthens to an excess of 160 m s^{-1} . The southern hemisphere jet is centered between 60°S and 30°S and between 100 Pa and 3 Pa, slightly more compressed than the early-spring jet in the opposing hemisphere (previous panel). By solstice ($L_s=90^\circ$; northern summer/southern winter), the southern hemisphere jet descends and weakens to 120 m s^{-1} . At the highest altitudes, above 1 Pa, a second jet begins to form, potentially the lowest portion of the thermospheric jet observed in Lellouch et al. (2019), similar to the jet we see at $L_s=300^\circ$.

The behavior of the jet is largely symmetric across hemispheres (Fig. 10), so the behavior of the simulated southern hemisphere winds in Fig. 9 over the remainder of the Titan year can be inferred from the progression of the winds of the opposite hemisphere. A clear trend easily visible in Fig. 10 is the rapid strengthening of the polar jet in early autumn followed by a weakening in mid winter and an additional period of strengthening in late winter and early spring, before

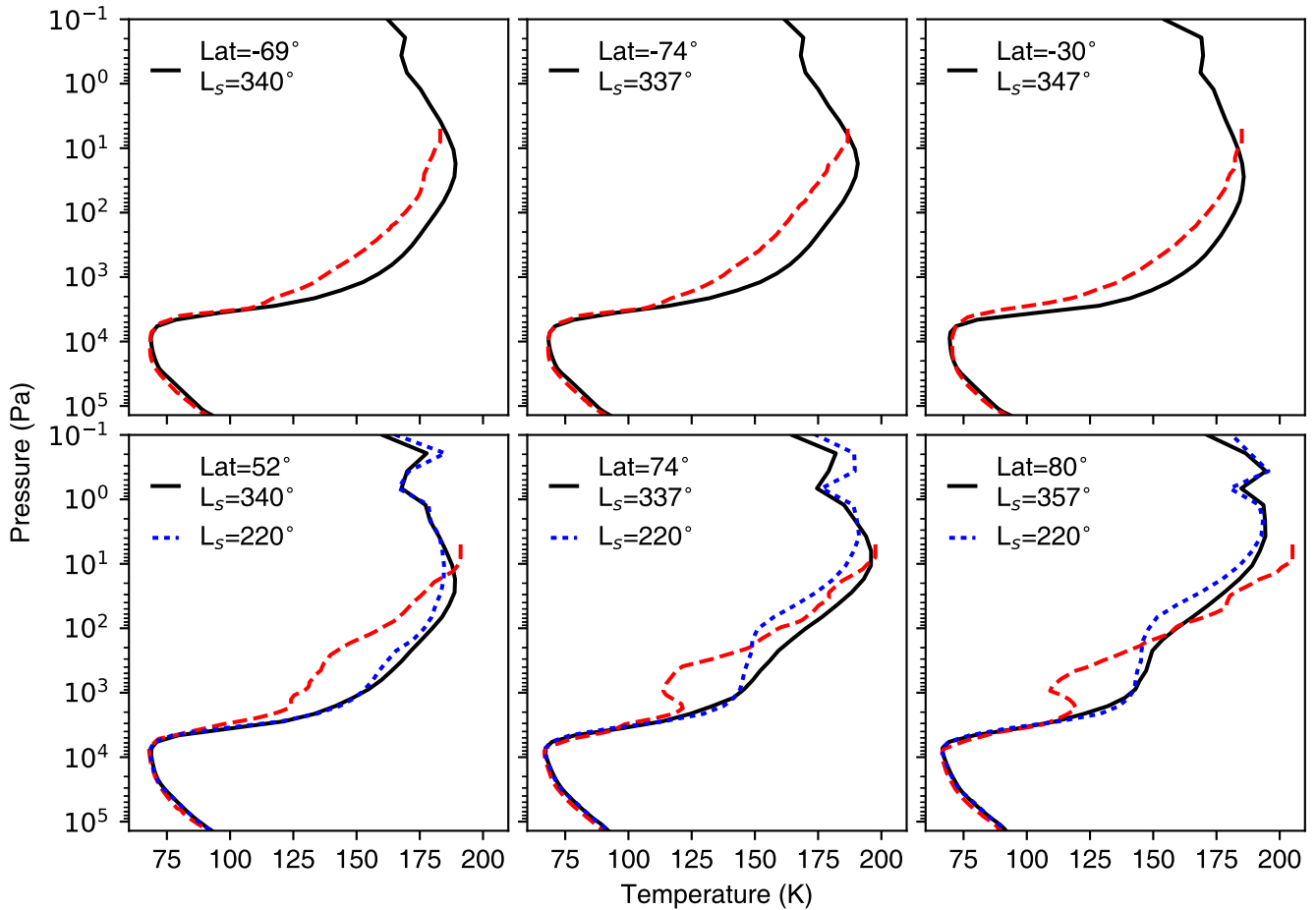


Fig. 6. Simulated zonal-mean temperature profiles (black) compared to radio occultation profiles (Schinder et al., 2012) (red dashed), and to model profiles from corresponding latitudes one full season prior to the timing of the radio occultations (blue dotted). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the jet dissipates in mid spring. This two-lobed structure is seen in observations from CIRS (Fig. 13 in Sharkey et al., 2021), though our simulated polar vortex reaches a maximum zonal wind $40\text{--}50^\circ L_s$ after the equinox, whereas the observations indicated a peak strength at $60^\circ L_s$ after equinox. This is consistent with an apparent shift in the temperature minima between our simulations and the observations. Our temperatures reach a minimum about $40^\circ L_s$ after equinox, whereas the observations in Teanby et al. (2019) show a minimum occurring about $50\text{--}60^\circ L_s$ after equinox.

The seasonal evolution of zonal winds at four pressure levels is shown in Fig. 10. Immediately apparent is the seasonality of the zonal wind (discussed previously) and the seasonal dependence of the effects caused by inclusion of SVRS. At the highest levels (1 Pa), inclusion of SVRS leads to significantly stronger zonal winds throughout the center of the jet. Lower down, near 11 Pa, the general seasonality is similar, though with impacts from SVRS significantly stronger near the poles than near the equator. Deeper in the stratosphere, at pressures near 67 Pa, effects from SVRS are significantly diminished, with only moderate strengthening less than 10 m s^{-1} near the southern mid winter poles, and weak strengthening in the southern winter hemisphere. By 322 Pa, inclusion of SVRS has effectively no impact on the zonal circulation.

3.2.3. Meridional circulation

The simulated meridional mass stream functions in early northern winter, late winter, just before and just after equinox, and in early spring are shown in Fig. 11. In early winter ($L_s=315^\circ$), the ascending

branch of the MOC is located above the summer pole, visible in the figure as a clockwise stream function (contour), along with adiabatic cooling (blue) associated with the upward motion and adiabatic warming (red) associated with the downward motion. At this season, the stratosphere is nearly perfectly hemispherically anti-symmetric. By mid winter ($L_s=340^\circ$), the single cell structure of the MOC begins to evolve, and the ascending branch of the cell is centered at 40°S . At this time, there exist three defined regions of adiabatic heating: a region of positive heating poleward of 50°S , a region of negative heating (cooling) between 50°S and 20°N , and a region of positive heating poleward of 20°N . At equinox ($L_s=0^\circ$), the ascending branch of the MOC, and the associated region of adiabatic cooling, is centered nearly over the equator, though the stream function tilts slightly southward above 1000 Pa. By early spring ($L_s=20^\circ$), the ascending branch is centered above 50°N in the low stratosphere (1000 Pa), with an enhanced southward tilt. This leads to adiabatic heating at upper levels and adiabatic cooling at lower levels north of the ascending branch of the MOC at 50°N . Finally, by northern mid spring ($L_s=45^\circ$), the MOC returns to the single-cell state. Once again, there is nearly perfect hemispheric anti-symmetry in adiabatic heating, with cooling in the spring hemisphere and heating in the autumn hemisphere.

The transition of the ascending branch of the MOC from the southern hemisphere to the northern hemisphere begins in northern mid winter and lasts until the middle of northern spring, a total of 90° of L_s , or one full season. However, the circulation of the stratosphere only flows from equator to pole in both hemispheres for a short time

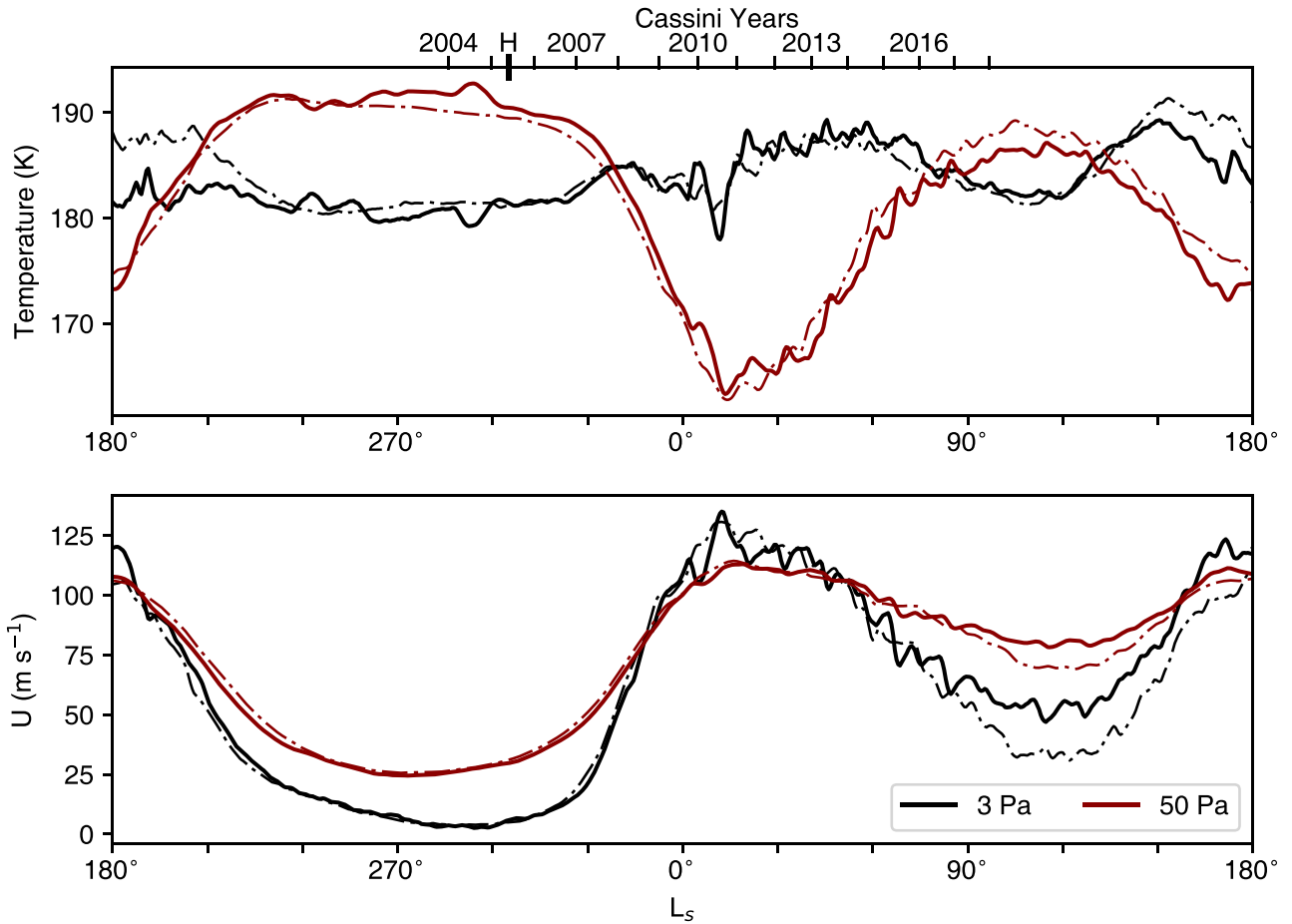


Fig. 7. Evolution of temperature (top panel) and zonal wind (bottom panel) averaged between 85°S and 60°S . Solid lines indicate the model run with SVRS and dot-dash lines indicate the model with static species. The 3 Pa level represents stratopause and the 50 Pa level represents the middle of the stratosphere. Timing of the Huygens descent is labeled H on the top axis.

around equinox; over much of the transition, the cross-equatorial cell dominates. [Vinatier et al. \(2015\)](#) (their Section 5.1) inferred the timing of the reversal of the MOC using stratospheric temperature and molecular abundance as proxies for the atmospheric dynamics. They began to observe a warming of the south polar mesosphere (above about 1 Pa) in January 2010 ($L_s=5^{\circ}$). At this time, they also observed an enrichment in aerosol abundance, presumably also an indicator of descending motion, though abundance profiles of trace species remained unaffected. Our simulation shows that the circulation should begin to reverse (and hence high altitude adiabatic warming should be present) as early as $L_s=340^{\circ}$ (July 2008). [Vinatier et al. \(2015\)](#) also suggested that two meridional overturning cells were present at $L_s=5^{\circ}$, which is consistent with what our simulations indicate. The MOC was inferred to exist in the two cell state until June 2010 ($L_s=10^{\circ}$), which is about 10° earlier than what our simulations predict. However, our simulations show that the lower stratospheric levels seem to precede the high levels as the MOC transitions from one state to the other, and that during the transition, the ascending branch is not perfectly vertical, but slants towards the originating hemisphere until the middle of spring. Therefore, the MOC never achieves perfect hemispheric symmetry.

The timing of the opposite transition, when the ascending branch moves from the northern to southern hemisphere, is very similar to the southern-to-northern transition in our simulations. The single cell structure begins to deteriorate around $L_s=140^{\circ}$ (40° before equinox), is approximately hemispherically symmetric at the equinox ($L_s=180^{\circ}$), and achieves a complete reversal by $L_s=220^{\circ}$ (40° after equinox). The N–S transition is about 10° of L_s shorter than the S–N transition, though

it still achieves approximate hemispheric symmetry at equinox. We do not have an explanation for the cause of the seasonal asymmetry of the MOC transition: a candidate explanation might invoke Saturn's 0.052 orbital eccentricity, but the fact that Saturn's perihelion occurs near $L_s=278^{\circ}$ would imply that the S–N transition should occur more quickly than the N–S transition, which is not the case in our simulations.

[Fig. 12](#) shows the effects of including SVRS on the meridional mass stream function (black contours) and the adiabatic heating rates (color shading). From early northern winter through mid northern spring there is strengthened south to north flow in the MOC. This results in enhanced flow during the early northern winter period and diminished flow in spring. An identical effect is seen in the other half of the year, during which time the north to south circulation is strengthened. The directionality of the strengthened flow reverses around each solstice.

Cassini measurements have previously been used to estimate the vertical velocity of air in the descending branch of the MOC. To estimate the vertical velocity (in mm s^{-1}) of the ascending and descending branches of the overturning circulation, we can multiply the adiabatic heating rates (calculated in the model as $RT\omega/P$, where R is the specific gas constant, T is the absolute temperature, ω is the pressure velocity, and P is the pressure) by $10^3 \times c_p/g$ (where c_p is the heat capacity at constant pressure and g is the gravitational acceleration; the product comes out to approximately 10^6 m K^{-1}). With this conversion, we can estimate the average vertical velocity of the overturning circulation near 10 Pa to be approximately 1 mm s^{-1} . Within the ascending and descending branches themselves, the magnitude of the vertical motion remains constant, regardless of the central latitude of

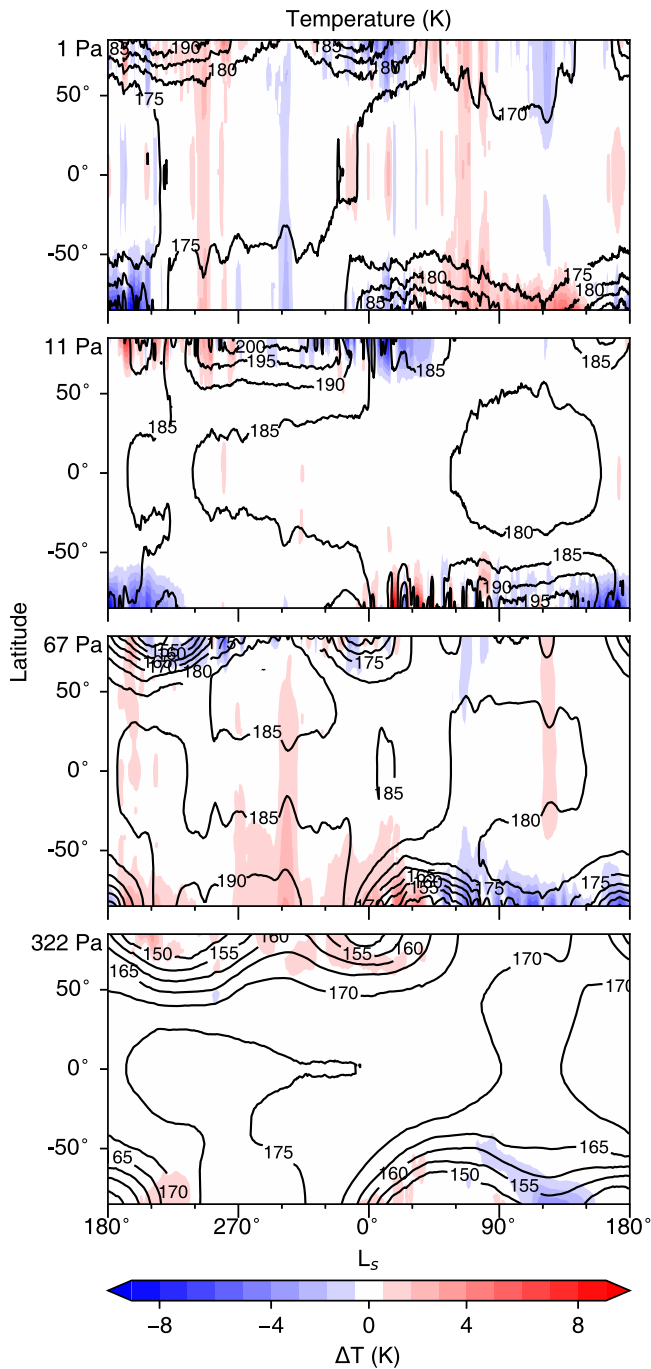


Fig. 8. Seasonal evolution of the stratospheric temperature at four pressure levels throughout Titan's stratosphere and lower mesosphere. Temperature contours are shown in black, and effects from inclusion of SVRS are shown in colors. Blue indicates relative cooling and red indicates relative warming caused by SVRS. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the branches; in other words, the simulated vertical velocity at 10 Pa in the ascending branch is of order 1 mm s^{-1} regardless of the season.

Teanby et al. (2012) used the adiabatic warming at the autumn stratopause to infer vertical velocities in the descending branch of the MOC. Above 300 km, they estimated subsidence velocities near 0.5 mm s^{-1} in early 2010 ($L_s=5^\circ$, the third plot of Fig. 11), that quickly increased to 1.5 mm s^{-1} by mid 2010 ($L_s=10^\circ$). Vinatier et al. (2015) used the descending molecular enrichment zone in southern autumn ($L_s=20\text{--}30^\circ$, corresponding to the fourth panel of Fig. 11)

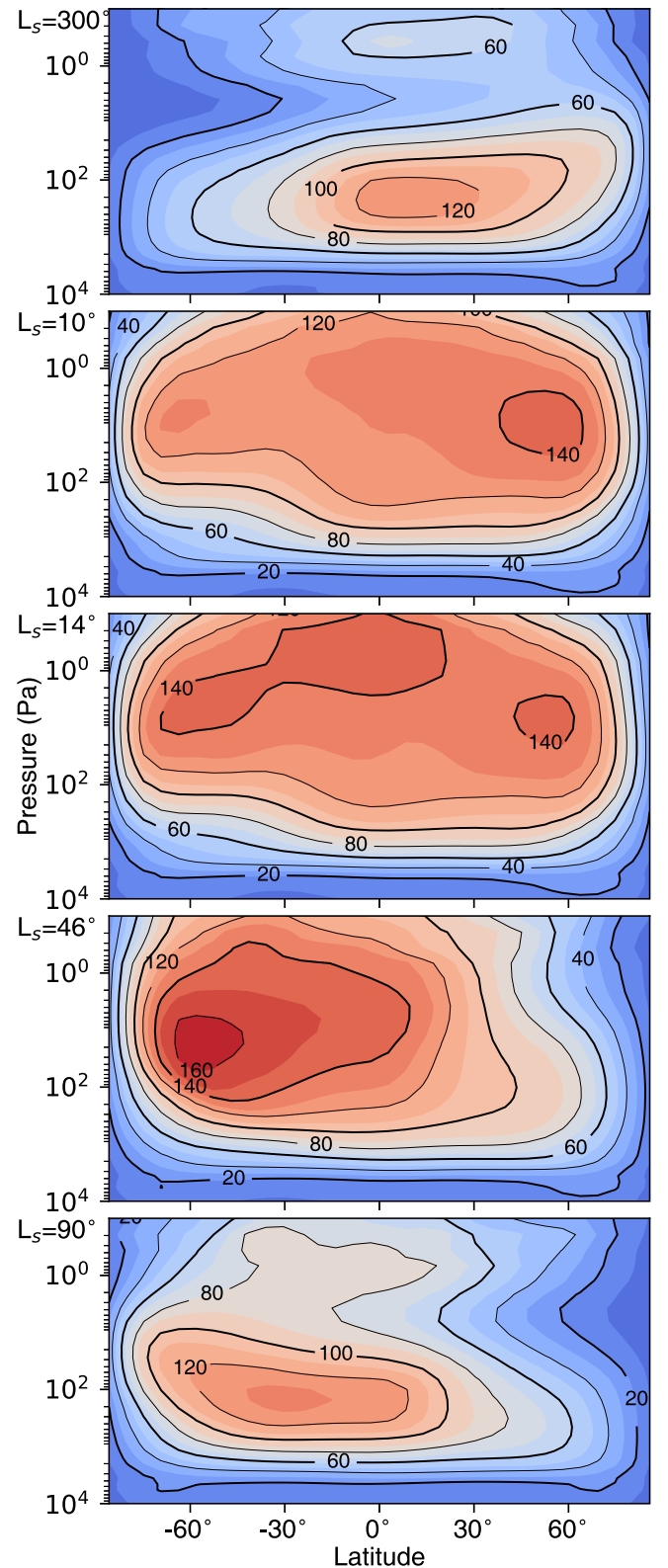


Fig. 9. Simulated stratospheric ($p < 10^4 \text{ Pa}$) zonal-mean zonal wind (m s^{-1}) during (from top to bottom) early northern (southern) winter (summer), early northern spring (autumn), early northern spring (autumn), mid northern spring (autumn), and northern summer (winter) solstice.

to infer the vertical velocity of the descending branch of the MOC above 1 Pa poleward of 80°S . At 450 km, they estimated a vertical

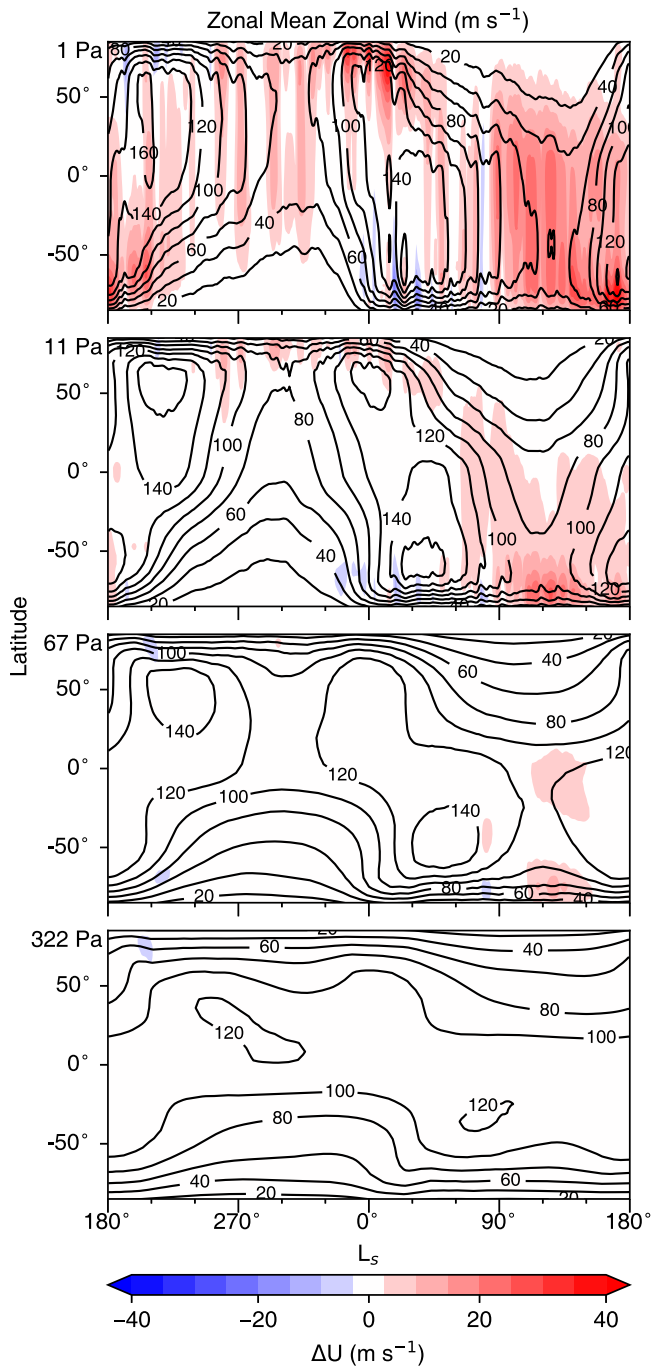


Fig. 10. Seasonal evolution of the zonal-mean zonal wind at four pressure levels over one full Titan year (black contours), and the effects from inclusion of SVRS (colored shading). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

velocity of 4.4 mm s^{-1} using HCN profiles and 12.0 mm s^{-1} using C_2H_2 profiles. More confidence was given to the HCN-derived velocity due to the steeper gradient in molecular abundance seen with that molecule. Deeper in the atmosphere, at 1 Pa or 380 km, a downward velocity of 2.5 mm s^{-1} was determined from HCN measurements. Our simulated vertical velocities are therefore generally consistent with these observational estimates.

3.2.4. Radiative heating

Fig. 13 shows the meridional cross-sections of zonal mean zonal wind, temperature, radiative heating due to aerosols, and radiative

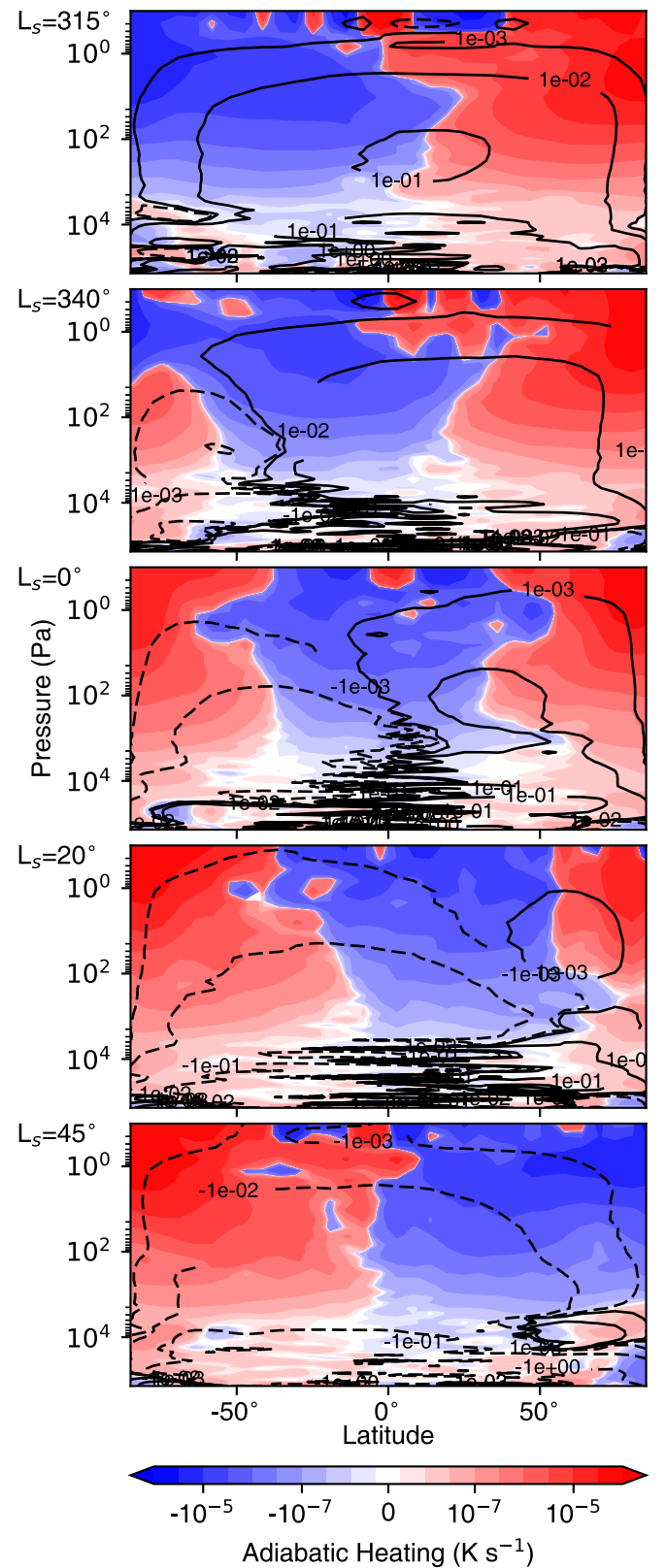


Fig. 11. The meridional mass stream function (10^9 kg s^{-1}) calculated in early northern winter, late northern winter, northern spring equinox, early northern spring, and mid northern spring. Negative values (dashed) indicate counter-clockwise circulation. Positive values (solid) indicate clockwise motion. Adiabatic heating (cooling) is shown by red (blue) shading and indicates descending (ascending) air. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

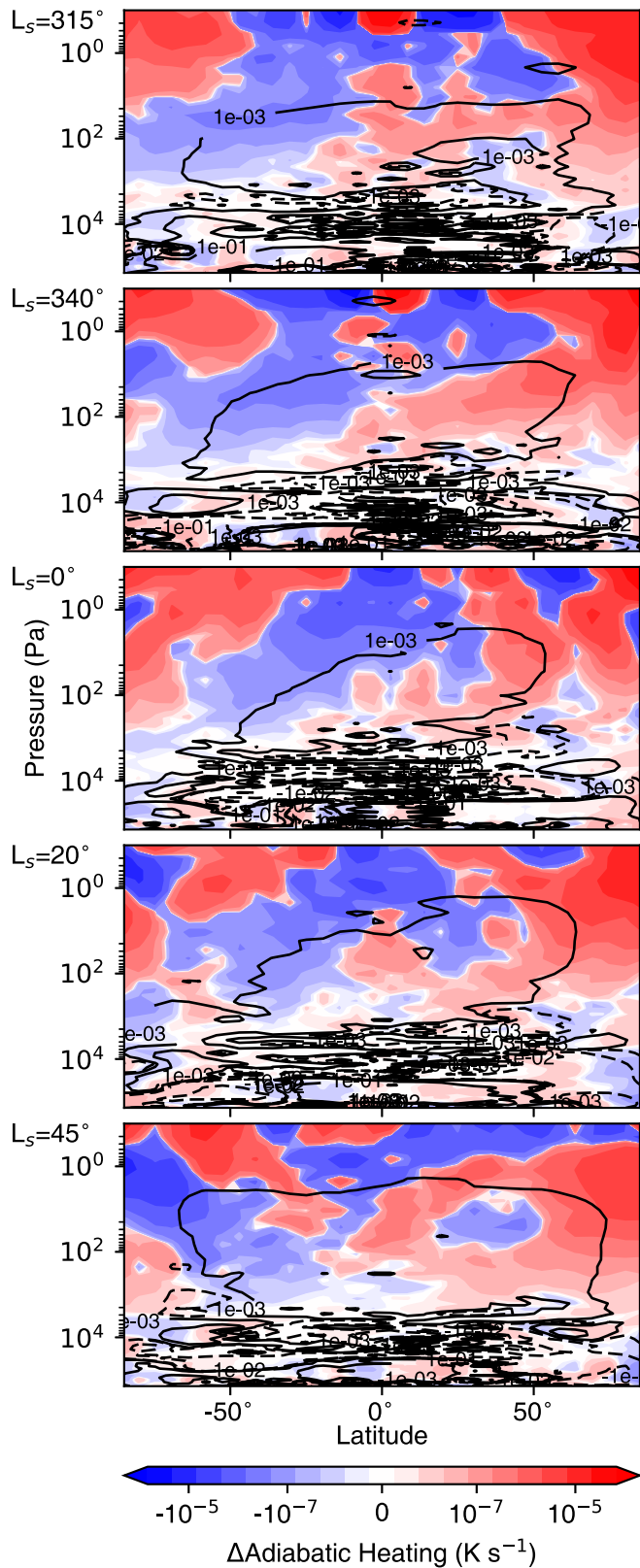


Fig. 12. Effects of the inclusion of SVRS on the meridional mass stream function (contours; 10^9 kg s^{-1}) and adiabatic heating rates (colors). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

heating due to molecular species during early northern winter, late northern winter, and mid northern spring. The individual rates due to either aerosols or gaseous molecules are calculated by setting the opacity of the other absorber equal to zero, and calculating the divergence of the resulting radiative flux. While the aerosol is the primary visible absorber, and while molecules represent the main thermal absorber, both contribute to both the shortwave and longwave radiative fluxes.

In early northern winter ($L_s = 300^\circ$), the stratospheric jet is strongest just north of the equator and vertically centered at 100 Pa with a maximum zonal wind above 120 m s^{-1} . Inclusion of SVRS leads to a strengthened upper polar flank near 20 Pa by 10 m s^{-1} , and strengthened zonal flow at all latitudes above 1 Pa. The stratopause temperature at most latitudes is about 180 K centered near 20 Pa. The exception to this is poleward of 60°N , where the stratopause warms to above 195 K at 10 Pa. The aerosol radiative heating in the stratosphere is positive southward (towards the summer hemisphere) of 30°N and negative northward of that latitude. In the mesosphere, above 0.5 Pa, the aerosol radiative heating is positive only southward of 60°S . The maximum aerosol radiative heating is centered at 10 Pa above the summer pole. This is likely due to the prolonged heating of the aerosols by solar visible light during the extended summer day. Including SVRS leads to decreased heating in the summer latitudes above 20 Pa, increased heating in the summer latitudes below 20 Pa, enhanced cooling at high winter polar latitudes, and weakened cooling at high altitudes above the equator. The molecular radiative heating is negative at all latitudes, with SVRS-induced stronger cooling rates at the winter pole and diminished cooling at all other latitudes. The strengthened radiative cooling in the upper atmosphere can be explained by the increased molecular abundance leading to increased thermal emission.

By the end of the northern winter (second column), the zonal winds strengthen and the vertical extent of the jet extends to above 10 Pa. Inclusion of SVRS leads to a strengthened poleward upper flank of the jet, as discussed above. The stratopause temperature remains the same from mid winter, but it decreases in pressure to 3 Pa above the winter pole, with SVRS acting to cool the stratopause by more than 6 K. The aerosol radiative heating is positive all the way to 60°N , indicative of the insolation approaching a more hemispherically symmetric state as Titan nears equinox. Including SVRS leads to opposite affects as in early winter, with aerosol radiative cooling diminished above the winter pole. The molecular radiative heating near equinox remains negative at all latitudes, though the meridional structure begins to be more symmetric. SVRS leads to diminished radiative cooling rates throughout most of the stratosphere, except at polar latitudes above 1 Pa, and at some lower levels above the northern winter pole.

In mid northern spring, the jet strengthens and moves to the opposite (autumn southern) hemisphere, centered at 60°S at 10 Pa. The jet extends higher in the spring hemisphere due to SVRS. The temperature structure is similar to the structure near equinox, though with the autumn stratopause both colder and higher than the spring stratopause. Inclusion of SVRS leads to cooler temperatures in both hemispheres above 100 Pa poleward of 60° . Aerosol radiative heating is similar to that simulated in mid winter, though reversed meridionally. Aerosol heating is positive northward of 30°S and negative southward of that. SVRS leads to enhanced heating rates in the spring hemisphere and diminished cooling above the autumn pole. Molecular heating is again negative at all latitudes (except the high altitude equatorial region), with spring cooling enhanced by SVRS, due to the residual enrichment in trace species.

4. Discussion

As described above, inclusion of SVRS affects both the mean state and the strength of seasonal cycle of the simulated atmosphere. Including SVRS leads to a generally cooler winter and autumn stratopause, likely due to the enrichment of radiatively active trace species. Similarly, Titan's zonal wind field experiences an enhanced seasonal cycle,

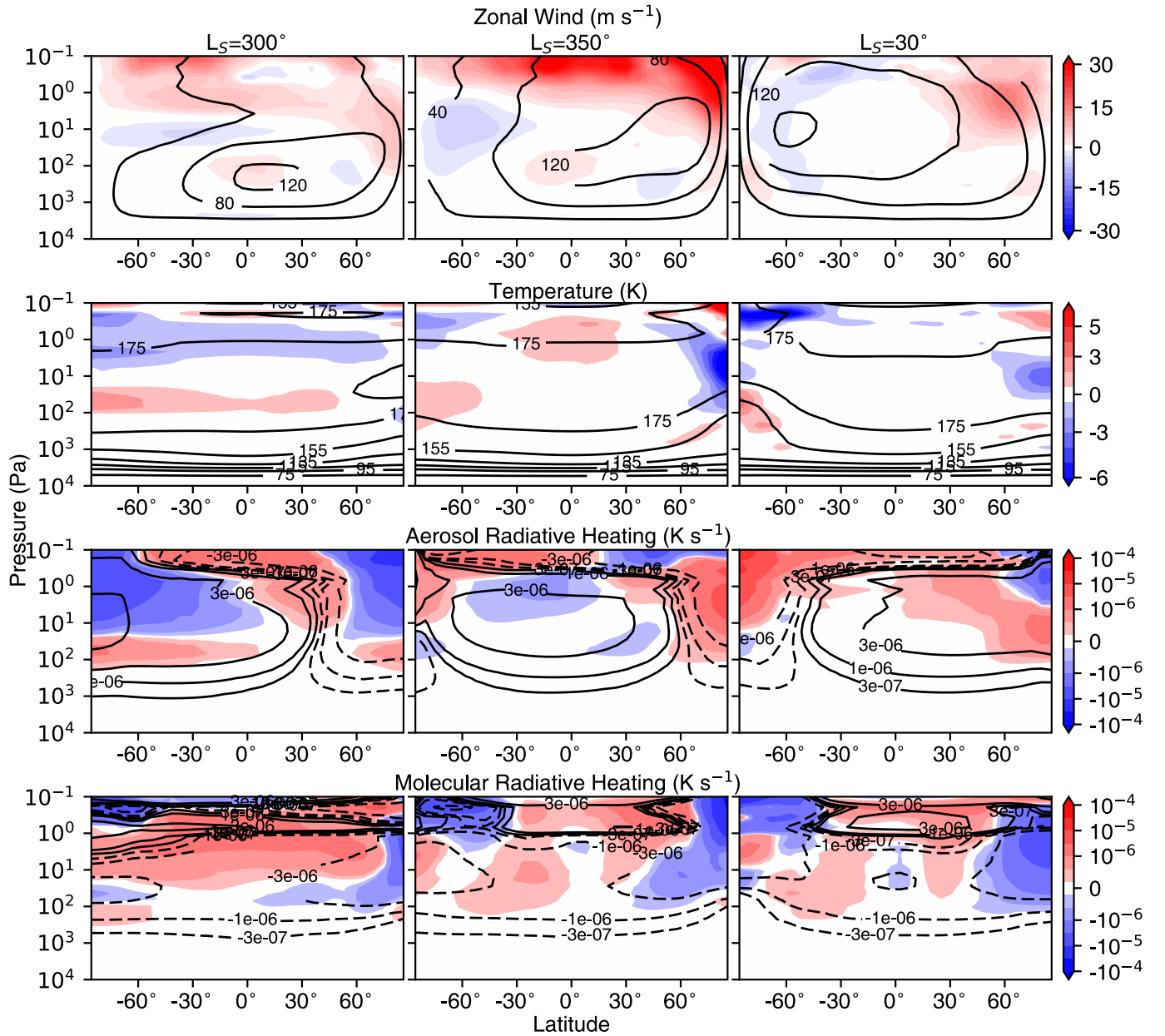


Fig. 13. Zonal-mean zonal wind, temperature, radiative heating due to aerosols, and the radiative heating due to molecular emission (contours) and the effect on each field from SVRS (shading) during early northern winter, late northern winter, and mid northern spring. The color scale for the bottom two rows are symmetric logarithmic scales, which are logarithmic at magnitudes greater than 10^{-7} and linear at lesser magnitudes. The effects of SVRS at pressures above 10^4 Pa (below the lower stratosphere) are negligible and not shown here. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with the jet typically strengthened at high levels throughout the year, and little-to-no strengthening in mid summer. Enhancement due to SVRS is strongest at higher altitudes. Similar to previous TAM simulations (Lora et al., 2015), the simulated stratosphere is warmer than observed with CIRS (Achterberg et al., 2008), even while including SVRS. Conversely, the zonal wind speed in autumn and spring is greater when SVRS is included, especially at pressures below 50 Pa; this brings the simulated winds closer to the speeds inferred from meridional temperature gradients (Sharkey et al., 2021), although they are still several tens of m s^{-1} too slow. While SVRS increases the magnitude of the seasonality of the temperature and zonal wind, it does not affect the timing of Titan's atmospheric phenomena, including the atmospheric temperature wave seen in radio occultations or the onset and dissipation of the polar vortex.

Our model shows an improved simulation of stratospheric temperatures compared to several previous models, though our zonal winds

are weaker in some cases. Specifically, the temperatures in our upper stratospheric, stratopause, and lower mesospheric levels are cooler and therefore closer to observations than those simulated by Crespin et al. (2008), Newman et al. (2011), and Lebonnois et al. (2012). The magnitude of winds in our simulation is similar to that simulated by Lebonnois et al. (2012); additionally, our model simulates an increasing wind shear above 1 Pa indicative of a mesospheric jet, which was also simulated by Lebonnois et al. (2012). On the other hand, the peak wind speeds in our simulation are about 40 m s^{-1} slower than those simulated by Crespin et al. (2008) and Newman et al. (2011), though we accurately simulate the observed mid winter weakening of the zonal winds described in Shultis et al. (2022).

Despite the stronger seasonal cycle driven by SVRS being an improvement from the static profile case used in previous versions of TAM and other GCMs, several potential sources of remaining model biases exist. First, a potentially significant source of discrepancy between the

simulated zonal winds of Titan GCMs and measured zonal winds was discussed in Tokano (2013). The centrifugal force of Titan's superrotating zonal winds causes an equatorial bulge, which is neglected in GCMs that rely on the hydrostatic approximation of vertical momentum (along with the shallow atmosphere approximation, which loses accuracy above Titan's stratopause). The equatorial bulge would increase the geopotential at low latitudes more than at high latitudes, which would then increase the geopotential gradient from equator to pole, which, by gradient balance, would then increase the zonal wind shear and zonal winds at high altitudes. Tokano (2013) studied the effect of including the vertical momentum from Titan's equatorial bulge in a quasi-hydrostatic GCM and found that inclusion of the centrifugal terms in the vertical momentum equations led to simulated zonal winds several tens of m s^{-1} faster than in the hydrostatic case. Along similar lines of intrinsic model discrepancies is the one-dimensional radiative transfer implementation. While this assumption is valid at low latitudes, as the solar zenith angle increases near the winter polar latitudes, incident shortwave light is affected by Titan's atmosphere at lower latitudes. While this is a problem in the mid-latitudes, it is resolved during polar night as no shortwave light reaches the highest polar latitudes. Thus, while the one dimensional radiative transfer approximation has intrinsic errors, they are not significant near winter polar night.

An additional source of discrepancies could be the lack of certain parameterizations of physical processes. Importantly, the species included in SVRS do not interact with the general circulation of the simulated atmosphere. This precludes radiative-dynamical coupling and feedbacks, as well as the possibility of small scale effects from localized radiative forcing. Still, such effects would likely only be relevant above the upper stratosphere, as radiative time constants there are significantly shorter than at lower altitudes, and our results already indicate only modest influences from strong compositional variations. Several previous models (Crespin et al., 2008; Lebonnois et al., 2012; Larson et al., 2015) have included a haze microphysics scheme that was coupled to their model dynamics. Interestingly, our model and that of Newman et al. (2011) (both of which lack haze microphysics) are unable to produce the zonal wind minimum around 2000 Pa (Bird et al., 2005). Though this is not necessarily caused by inclusion of the coupled haze microphysics, Crespin et al. (2008) and Lebonnois et al. (2012) were both able to simulate a local zonal wind minimum at around the same pressure. While including SVRS accounts for aerosol variations in the upper stratosphere caused by the dynamic DHL, it does not include small scale variations that could exist lower down, where the zonal wind minimum was seen.

Another unmodeled physical process that could affect the stratosphere is condensation from trace species. In the descending portions of the MOC, trace molecules are delivered to lower altitudes and lower temperatures, which could cause them to condense. The latent heat of condensation release by this process would warm the lower stratosphere at high latitudes. Condensation was studied in the two-dimensional, axisymmetric model of Hourdin and Lebonnois (2004). That model, however, sought to study the cause of the meridional structure of Titan's trace species, and not how the latent heat of condensation affects Titan's general circulation, and it is unclear how they treated latent heat released by this process.

While condensing species release latent heat, they may also form clouds. Several types of stratospheric clouds have been observed on Titan, including HCN (De Kok et al., 2014; West et al., 2016) and C_2H_6 (Griffith et al., 2006). The presence of clouds in Titan's stratosphere would affect Titan's radiative balance and therefore temperature structure. If the clouds have a net cooling effect in the autumn and winter polar regions, this could cause an increased vertical wind shear (due to gradient wind balance) leading to increased stratospheric zonal winds. No previous Titan GCMs have included the radiative effects of stratospheric clouds.

Finally, to increase the stability of the model, two assumptions were included at the top-most layers. The first is a diffusive sponge, used in previous versions of TAM (Lora et al., 2015), that acts on the zonal mean zonal wind, zonal mean meridional wind, and eddy components of the wind at the top layer. A second modification is the inclusion of Rayleigh drag on the top three model levels. The Rayleigh drag damps the zonal and meridional eddy velocity components and the temperature eddy component towards the zonal mean of each, respectively. The damping coefficient is strongest at the lowest pressure, and decreases as pressure increases. The Rayleigh drag serves to reduce the effects of wave reflection at the top layer and reduces the overall amplitude of vertically propagating waves, which tend to increase with altitude and destabilize the model. While inclusion of both of these effects has been necessary for successful model runs, they may also serve as a sink for atmospheric momentum, and reduce the realism of the top levels of the model. The Rayleigh drag only acts down to 1 Pa in the simulations presented here (the top three layers of the model), so levels at higher pressures should be reliable.

5. Conclusion

We have developed a dataset of seasonally varying trace molecule abundance and aerosol opacity for Titan's atmosphere derived from Cassini visible and infrared observations (referred to as the SVRS dataset). Using this new product, we have studied the effects of seasonal variations in radiatively active species on Titan's atmospheric circulation with TAM.

The SVRS dataset was developed by combining nadir observations of Titan's trace molecules from Teanby et al. (2019) with limb measurements from Mathé et al. (2020), as well as combining vertical-meridional cross-sections of aerosol opacity from Seignovert et al. (2021) with in-situ measurements at lower altitudes from Dooze et al. (2016). The SVRS dataset provides a single resource which captures information on the observed seasonality of the meridional and vertical distributions of C_2H_6 , C_2H_4 , C_2H_2 , C_3H_4 , C_4H_2 , HCN, and HC_3N , as well as aerosol opacity. Trace molecules tend to be enriched above the poles beginning in early autumn, and remain enriched through the following spring. Aerosol opacity profiles are constant below 100 Pa, and above this level SVRS includes seasonal variations in the evolution of the detached haze layer from its formation in summer through its descent in autumn.

Inclusion of SVRS in TAM leads to a strengthened polar vortex in autumn at levels above 50 Pa (the upper stratosphere into the stratopause), with zonal winds up to 40 m s^{-1} faster above 10 Pa. The enhanced flow follows the jet center, and diminishes by the mid stratosphere. Though inclusion of seasonally varying radiatively active species does lead to a cooler stratopause in winter, we find that the timing of temperature variations does not depend sensitively on the seasonality of the radiative species; temperature variations occur at similar pacing in both the simulation with SVRS and in the control simulation with static species. Radiative heating rates due to Titan's aerosols tend to be weakened in the mid stratosphere, except near the early autumn pole. Similarly, radiative heating from trace molecules tends to be weaker in the mid stratosphere and stronger above 10 Pa. Our results suggest that models of Titan's atmosphere should include such seasonal variations for more robust simulations, as these variations lead to several changes in Titan's atmospheric state across the Titan year.

Data availability

The SVRS dataset is publically available on <https://doi.org/10.5281/zenodo.7222719>. The archive also includes the zonally averaged zonal wind, meridional wind, temperature, and meridional mass stream function as simulated in TAM with SVRS radiative forcing.

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References

- Achterberg, R.K., Conrath, B.J., Gierasch, P.J., Flasar, F.M., Nixon, C.A., 2008. Titan's middle-atmospheric temperatures and dynamics observed by the Cassini Composite Infrared Spectrometer. *Icarus* 194, <http://dx.doi.org/10.1016/j.icarus.2007.09.029>.
- Achterberg, R.K., Gierasch, P.J., Conrath, B.J., Flasar, F.M., Nixon, C.A., 2011. Temporal variations of Titan's middle-atmospheric temperatures from 2004 to 2009 observed by Cassini/CIRS. *Icarus* 211, <http://dx.doi.org/10.1016/j.icarus.2010.08.009>.
- Achterberg, R.K., Gierasch, P., Contrath, B., Flasar, F., Jennings, D.E., Nixon, C.A., 2014. Post-equinox Variations of Titan's mid-stratospheric temperatures from Cassini/CIRS observations. In: AAS/Division for Planetary Sciences Meeting Abstracts. 46, p. 102.07.
- Anderson, C.M., Samuelson, R.E., 2011. Titan's aerosol and stratospheric ice opacities between 18 and 500 μm : Vertical and spectral characteristics from Cassini CIRS. *Icarus* 212 (2), 762–778. <http://dx.doi.org/10.1016/j.icarus.2011.01.024>.
- Battalio, J.M., Lora, J.M., Rafkin, S., Soto, A., 2022. The interaction of deep convection with the general circulation in Titan's atmosphere. Part 2: Impacts on the climate. *Icarus* 373, 114623.
- Bézard, B., Vinatier, S., Achterberg, R.K., 2018. Seasonal radiative modeling of Titan's stratospheric temperatures at low latitudes. *Icarus* 302, 437–450. <http://dx.doi.org/10.1016/j.icarus.2017.11.034>.
- Bird, M., Allison, M., Asmar, S., Atkinson, D., Avruch, I., Dutta-Roy, R., Dziemia, Y., Edenhofer, P., Folkner, W., Gurvits, L., et al., 2005. The vertical profile of winds on Titan. *Nature* 438 (7069), 800–802.
- Coustonis, A., Jennings, D.E., Achterberg, R.K., Bampasidis, G., Nixon, C.A., Lavvas, P., Cottini, V., Flasar, F.M., 2018. Seasonal evolution of Titan's stratosphere near the poles. *Astrophys. J.* 854 (2), L30. <http://dx.doi.org/10.3847/2041-8213/aaadbd>.
- Crespin, A., Lebonnois, S., Vinatier, S., Bézard, B., Coustenis, A., Teanby, N.A., Achterberg, R.K., Rannou, P., Hourdin, F., 2008. Diagnostics of Titan's stratospheric dynamics using Cassini/CIRS data and the 2-dimensional IPSL circulation model. *Icarus* 197, 556–571.
- De Kok, R.J., Teanby, N.A., Maltagliati, L., Irwin, P.G., Vinatier, S., 2014. HCN ice in Titan's high-altitude southern polar cloud. *Nature* 514 (7520), 65–67.
- Doose, L.R., Karkoschka, E., Tomasko, M.G., Anderson, C.M., 2016. Vertical structure and optical properties of Titan's aerosols from radiance measurements made inside and outside the atmosphere. *Icarus* 270, 355–375. <http://dx.doi.org/10.1016/j.icarus.2015.09.039>.
- Faulk, S.P., Lora, J.M., Mitchell, J.L., Milly, P.C.D., 2020. Titan's climate patterns and surface methane distribution due to the coupling of land hydrology and atmosphere. *Nat. Astron.* 4 (4), 390–398. <http://dx.doi.org/10.1038/s41550-019-0963-0>.
- Flasar, F.M., Achterberg, R.K., 2009. The structure and dynamics of Titan's middle atmosphere. *Phil. Trans. R. Soc. A* 367, 649–664.
- Flasar, F., Achterberg, R., Conrath, B., Gierasch, P., Kunde, V., Nixon, C., Bjoraker, G., Jennings, D., Romani, P., Simon-Miller, A., et al., 2005. Titan's atmospheric temperatures, winds, and composition. *Science* 308 (5724), 975–978.
- Flasar, F.M., Samuelson, R.E., Conrath, B.J., 1981. Titan's atmosphere: temperature and dynamics. *Nature* 292, 696–698.
- Friedson, A.J., West, R.A., Wilson, E.H., Oyafuso, F., Orton, G.S., 2009. A global climate model of Titan's atmosphere and surface. *Planet. Space Sci.* 57 (14–15), 1931–1949.
- Gordon, I., Rothman, L., Hill, C., Kochanov, R., Tan, Y., Bernath, P., Birk, M., Boudon, V., Campargue, A., Chance, K., Drouin, B., Flaud, J.-M., Gamache, R., Hodges, J., Jacquemart, D., Perevalov, V., Perrin, A., Shine, K., Smith, M.-A., Tennyson, J., Toon, G., Tran, H., Tyuterev, V., Barbe, A., Császár, A., Devi, V., Furtenbacher, T., Harrison, J., Hartmann, J.-M., Jolly, A., Johnson, T., Karman, T., Kleiner, I., Kyuberis, A., Loos, J., Lyulin, O., Massie, S., Mikhailenko, S., Moazzen-Ahmadi, N., Müller, H., Naumenko, O., Nikitin, A., Polyansky, O., Rey, M., Rotger, M., Sharpe, S., Sung, K., Starikova, E., Tashkun, S., Auwera, J.V., Wagner, G., Wilzewski, J., Wcislo, P., Yu, S., Zak, E., 2017. The HITRAN2012 spectroscopic database. *JQSRT* 203, 3–69.
- Gordon, C.T., Stern, W.F., 1982. A description of the GFDL global spectral model. *Mon. Weather Rev.* 110 (7), 625–644.
- Griffith, C.A., Penteado, P., Rannou, P., Brown, R., Boudon, V., Baines, K.H., Clark, R., Drossart, P., Buratti, B., Nicholson, P., McKay, C.P., Coustenis, A., Negrao, A., Jaumann, R., 2006. Evidence for a polar ethane cloud on Titan. *Science* 313, 1620–1622. <http://dx.doi.org/10.1126/science.1131867>.
- Harris, L., Chen, X., Putman, W., Zhou, L., Chen, J.-H., 2021. A scientific description of the GFDL finite-volume cubed-sphere dynamical core.
- Hayes, A.G., 2016. The lakes and seas of Titan. *Annu. Rev. Earth Planet. Sci.* 44 (1), 57–83. <http://dx.doi.org/10.1146/annurev-earth-060115-012247>.
- Hayes, A.G., Lorenz, R.D., Lunine, J.I., 2018. A post-Cassini view of Titan's methane-based hydrologic cycle. *Nat. Geosci.* 11 (5), 306–313. <http://dx.doi.org/10.1038/s41561-018-0103-y>.
- Hörst, S.M., 2017. Titan's atmosphere and climate. *J. Geophys. Res.: Planets* 122 (3), 432–482. <http://dx.doi.org/10.1002/2016je005240>.
- Hourdin, F., Lebonnois, S., 2004. Titan's stratospheric composition driven by condensation and dynamics. vol. 109, p. E12005. <http://dx.doi.org/10.1029/2004JE002282>.
- Hourdin, F., Talagrand, O., Sadourny, R., Courtin, R., Gautier, D., McKay, C.P., 1995. Numerical simulation of the general circulation of Titan. *Icarus* 117, 358–374.
- Jennings, D., Tokano, T., Cottini, V., Nixon, C., Achterberg, R., Flasar, F., Kunde, V., Romani, P., Samuelson, R., Segura, M., et al., 2019. Titan surface temperatures during the Cassini mission. *Astrophys. J. Lett.* 877 (1), L8.
- Kostiuk, T., Livengood, T.A., Sonnabend, G., Fast, K.E., Hewagama, T., Murakawa, K., Tokunaga, A.T., Annen, J., Buhl, D., Schmülling, F., Luz, D., Witaske, O., 2006. Stratospheric global winds on Titan at the time of Huygens descent. *J. Geophys. Res.: Planets* 111 (E07), <http://dx.doi.org/10.1029/2005JE002630>.
- Lacis, A., Oinas, V., 1991. A description of the correlated k distribution method for modeling nongray gaseous absorption, thermal emission, and multiple scattering in vertically inhomogeneous atmospheres. *J. Geophys. Res.* 96, 9027–9063.
- Larson, E.J., Toon, O.B., Friedson, A.J., 2014. Simulating Titan's aerosols in a three dimensional general circulation model. *Icarus* 243, 400–419.
- Larson, E.J., Toon, O.B., West, R.A., Friedson, A.J., 2015. Microphysical modeling of Titan's detached haze layer in a 3D GCM. *Icarus* 254, 122–134. <http://dx.doi.org/10.1016/j.icarus.2015.03.010>.
- Lavvas, P., Sander, M., Kraft, M., Imanaka, H., 2011. Surface chemistry and particle shape: processes for the evolution of aerosols in Titan's atmosphere. *Astrophys. J.* 728 (2), 80. <http://dx.doi.org/10.1088/0004-637x/728/2/80>.
- Lavvas, P., Yelle, R.V., Koskinen, T., Bazin, A., Vuitton, V., Vigren, E., Galand, M., Wellbrock, A., Coates, A.J., Wahlund, J.-E., Cray, F.J., Snowden, D., 2014. Aerosol growth in Titan's ionosphere. *Proc. Natl. Acad. Sci. USA* 110 (8), 2729–2734.
- Lebonnois, S., Burgalat, J., Rannou, P., Charnay, B., 2012. Titan global climate model: A new 3-dimensional version of the IPSL Titan GCM. *Icarus* 218 (1), 707–722. <http://dx.doi.org/10.1016/j.icarus.2011.11.032>.
- Lebonnois, S., Toubanc, D., Hourdin, F., Rannou, P., 2001. Seasonal variations of Titan's atmospheric composition. *Icarus* 152, 384–406.
- Lellouch, E., Gurwell, M., Moreno, R., Vinatier, S., Strobel, D., Moullet, A., Butler, B., Lara, L., Hidayat, T., Villard, E., 2019. An intense thermospheric jet on Titan. *Nat. Astron.* 3 (7), 614–619.
- Lombardo, N.A., Nixon, C.A., Achterberg, R.K., Jolly, A., Sung, K., Irwin, P.G.J., Flasar, F.M., 2019. Spatial and seasonal variations in C3H₈ hydrocarbon abundance in Titan's stratosphere from Cassini CIRS observations. *Icarus* 317, 454–469. <http://dx.doi.org/10.1016/j.icarus.2018.08.027>.
- Lora, J.M., Ádámkovics, M., 2017. The near-surface methane humidity on Titan. *Icarus* 286, 270–279. <http://dx.doi.org/10.1016/j.icarus.2016.10.012>.
- Lora, J.M., Battalio, J.M., Yap, M., Baciocco, C., 2022. Topographic and orbital forcing of Titan's hydroclimate. *Icarus* 115095.
- Lora, J.M., Lunine, J.I., Russell, J.L., 2015. GCM simulations of Titan's middle and lower atmosphere and comparison to observations. *Icarus* 250, 516–528. <http://dx.doi.org/10.1016/j.icarus.2014.12.030>.
- Lunine, J.I., Lorenz, R.D., 2009. Rivers, lakes, dunes, and rain: Crustal processes in Titan's methane cycle. *Annu. Rev. Earth Planet. Sci.* 37, 299–320.
- Mathé, C., Vinatier, S., Bézard, B., Lebonnois, S., Gorius, N., Jennings, D.E., Mamoutkine, A., Guandique, E., d'Ollone, J.V., 2020. Seasonal changes in the middle atmosphere of Titan from cassini/CIRS observations: Temperature and trace species abundance profiles from 2004 to 2017. *Icarus* 344, 113547. <http://dx.doi.org/10.1016/j.icarus.2019.113547>.
- Mitchell, J.L., Lora, J.M., 2016. The climate of Titan. *Annu. Rev. Earth Planet. Sci.* 44, 353–380.
- Newman, C.E., Lee, C., Lian, Y., Richardson, M.I., Toigo, A.D., 2011. Stratospheric superrotation in the TitanWRF model. *Icarus* 213 (2), 636–654. <http://dx.doi.org/10.1016/j.icarus.2011.03.025>.
- Nichols-Fleming, F., Corlies, P., Hayes, A.G., Ádámkovics, M., Rojo, P., Rodriguez, S., Turtle, E.P., Lora, J.M., Soderblom, J.M., 2021. Tracking short-term variations in the haze distribution of Titan's atmosphere with SINFONI VLT. *Planet. Sci. J.* 2 (5), 180.
- Niemann, H.B., Atreya, S.K., Demick, J.E., Gautier, D., Haberman, J.A., Harpold, D.N., Kasprzak, W.T., Lunine, J.I., Owen, T.C., Raulin, F., 2010. Composition of Titan's lower atmosphere and simple surface volatiles as measured by the Cassini-Huygens probe gas chromatograph mass spectrometer experiment. *J. Geophys. Res.* 115, E12006-E12028.
- Rannou, P., Cabane, M., Botet, R., Chassefiere, E., 1997. A new interpretation of scattered light measurements at Titan's limb. *J. Geophys. Res.* 105 (E5), 10977–11013.
- Rannou, P., Hourdin, F., McKay, C.P., 2002. A wind origin for Titan's haze structure. *Nature* 418, 853–856. <http://dx.doi.org/10.1038/nature00961>.
- Rannou, P., Hourdin, F., McKay, C., Luz, D., 2004. A coupled dynamics-microphysics model of Titan's atmosphere. *Icarus* 170 (2), 443–462.
- Schinder, P.J., Flasar, F.M., Marouf, E.A., French, R.G., McGhee, C.A., Kliore, A.J., Rappaport, N.J., Barbinis, E., Fleischman, D., Anabtawi, A., 2012. The structure of Titan's atmosphere from Cassini radio occultations: Occultations from the Prime and Equinox missions. *Icarus* 221 (2), 1020–1031. <http://dx.doi.org/10.1016/j.icarus.2012.10.021>.

- Seignovert, B., Rannou, P., West, R.A., Vinatier, S., 2021. Haze seasonal variations of Titan's upper atmosphere during the Cassini mission. *Astrophys. J.* 907 (1), 36. <http://dx.doi.org/10.3847/1538-4357/abcd3b>.
- Sharkey, J., Teanby, N.A., Sylvestre, M., Mitchell, D.M., Seviour, W.J., Nixon, C.A., Irwin, P.G., 2021. Potential vorticity structure of Titan's polar vortices from Cassini CIRS observations. *Icarus* 354, 114030. <http://dx.doi.org/10.1016/j.icarus.2020.114030>.
- Shultis, J., Waugh, D., Toigo, A., Newman, C., Teanby, N.A., Sharkey, J., 2022. Winter weakening of Titan's stratospheric polar vortices. *The Planetary Science Journal* 3 (4), 73. <http://dx.doi.org/10.3847/PSJ/ac5ea1>.
- Teanby, N.A., Bézard, B., Vinatier, S., Sylvestre, M., Nixon, C.A., Irwin, P.G.J., de Kok, R.J., Calcutt, S.B., Flasar, F.M., 2017. The formation and evolution of Titan's winter polar vortex. *Nature Commun.* 8 (1), <http://dx.doi.org/10.1038/s41467-017-01839-z>.
- Teanby, N., Irwin, P., de Kok, R., Jolly, A., Bézard, B., Nixon, C., Calcutt, S., 2009. Titan's stratospheric C₂N₂, C₃H₄, and C₄H₂ abundances from Cassini/CIRS far-infrared spectra. *Icarus* 202, 620–631.
- Teanby, N.A., de Kok, R., Irwin, P.G.J., Osprey, S., Vinatier, S., Gierasch, P.J., Read, P.L., Flasar, F.M., Conrath, B.J., Achterberg, R.K., Bézard, B., Nixon, C.A., Calcutt, S.B., 2008. Titan's winter polar vortex structure revealed by chemical tracers. *J. Geophys. Res.* 113 (E12), <http://dx.doi.org/10.1029/2008je003218>.
- Teanby, N.A., Patrick G. J. Irwin and, C.A.N., de Kok, R., Vinatier, S., Coustenis, A., Sefton-Nash, E., Calcutt, S.B., Flasar, F.M., 2012. Active upper-atmosphere chemistry and dynamics from polar circulation reversal on Titan. *Nature* 491, 732–735.
- Teanby, N.A., Sylvestre, M., Sharkey, J., Nixon, C.A., Vinatier, S., Irwin, P.G.J., 2019. Seasonal evolution of Titan's stratosphere during the Cassini mission. *Geophys. Res. Lett.* 46 (6), 3079–3089. <http://dx.doi.org/10.1029/2018gl081401>.
- Tokano, T., 2013. Wind-induced equatorial bulge in Venus and Titan general circulation models: Implication for the simulation of superrotation. *Geophys. Res. Lett.* 40 (17), 4538–4543.
- Tomasko, M.G., Bézard, B., Dose, L., Engel, S., Karkoschka, E., Vinatier, S., 2008. Heat balance in Titan's atmosphere. *Planet. Space Sci.* 56, 648–659.
- Vinatier, S., Bézard, B., Fouchet, T., Teanby, N., de Kok, R., Irwin, P.G.J., Conrath, B.J., Romani, P.N., Flasar, F.M., Coustenis, A., 2007. Vertical abundance profiles of hydrocarbons in Titan's atmosphere at 15S and 80N retrieved from Cassini/CIRS spectra. *Icarus* 188, 120–138.
- Vinatier, S., Bézard, B., de Kok, R., Anderson, C.M., Samuelson, R.E., Nixon, C.A., Mamoutkine, A., Carlson, R.C., Jennings, D.E., Guandique, E.A., 2010. Analysis of Cassini/CIRS limb spectra of Titan acquired during the nominal mission II: Aerosol extinction profiles in the 600–1420 cm⁻¹ spectral range. *Icarus* 210 (2), 852–866. <http://dx.doi.org/10.1016/j.icarus.2010.06.024>.
- Vinatier, S., Bézard, B., Lebonnois, S., Teanby, N.A., Achterberg, R.K., Gorius, N., Mamoutkine, A., Guandique, E., Jolly, A., Jennings, D.E., Flasar, F.M., 2015. Seasonal variations in Titan's middle atmosphere during the northern spring derived from Cassini / CIRS observations. *Icarus* 250, 95–115.
- West, R.A., Balloch, J., Dumont, P., Lavvas, P., Lorenz, R., Rannou, P., Ray, T., Turtle, E.P., 2011. The evolution of Titan's detached haze layer near equinox in 2009. *Geophys. Res. Lett.* 38 (6).
- West, R.A., Del Genio, A.D., Barbara, J.M., Toledo, D., Lavvas, P., Rannou, P., Turtle, E.P., Perry, J., 2016. Cassini Imaging Science Subsystem observations of Titan's south polar cloud. *Icarus* 270, 399–408.
- West, R.A., Seignovert, B., Rannou, P., Dumont, P., Turtle, E.P., Perry, J., Roy, M., Ovanessian, A., 2018. The seasonal cycle of Titan's detached haze. *Nat. Astron.* 2 (6), 495–500. <http://dx.doi.org/10.1038/s41550-018-0434-z>.